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Supplementary Table 1 Basic information of the consulted experts (n=15)

Item	Classification	Number	Percentage
Sex	Female	8	53.3
	Male	7	46.7
Age	30-45years old	12	80.0
	46-55years old	3	20.0
Education level	Bachelor	7	46.7
	Master's degree or above	8	53.3
Professional titles	Physician or nurse-in-charge	7	46.7
	associate chief physician or chief physician or co-chief superintendent nurse or chief superintendent nurse	8	53.3
Positional duties	Clinical education leader	4	26.7
	Head nurse	3	20.0
	Associate professor	4	26.7
	professor	4	26.7
Specialist work experience	10-20 years	12	80.0
	21-30 years	3	20.0

Supplementary Table 2 Variables for Risk Factors of PPCs After Lung Cancer Surgery: Results of the First Round of Delphi Consultation

Primary indicators	NO	Secondary indicators	$\overline{\chi} \pm S$	Coefficient of Variation
A. Demographic indicators	1	Age	4.80±0.56	0.12
	2	Sex	4.60±0.51	0.11
	3	Smoking history	5.00±0.00	0.00
	4	Body mass index (BMI)	4.67±0.72	0.16
	5	The length of hospitalization (Days)	4.27±0.88	0.21
	6	Living environment (urban, rural area)	4.00±1.20	0.30
	7	Marriage	2.20±1.27	0.58
	8	Occupation	3.40±1.45	0.43
	9	Lenth of hospital stay	2.40±1.55	0.65
	10	Education level	3.07±1.53	0.50
	11	The day of admission	1.93±1.34	0.69
	12	The day of surgery	3.13±1.69	0.54
	13	Smoking years	4.27±1.53	0.36
	14	Alcohol history (volume)	3.07±1.10	0.36
	15	Alcohol history (years)	3.13±1.13	0.36
	16	Primary diagnosis	3.60±1.45	0.40
	17	Secondly diagnosis	3.00±1.20	0.40
	18	Thirdly diagnosis	3.00±1.20	0.40
	19	Forthly diagnosis	3.07±1.16	0.38
	20	Family history of lung cancer	4.27±1.16	0.27
	21	History of radiotherapy and chemotherapy	4.93±0.26	0.05
	22	Preoperative pulmonary rehabilitation	4.73±0.59	0.13
	23	Type of chemotherapeutics	3.07±1.58	0.51
	24	History of chest surgery	4.73±0.59	0.13
	25	Pulmonary rehabilitation interventions	3.80±1.37	0.36
	26	Antibiotics administered before surgery	3.73±1.39	0.37
	27	The name and dose of preoperative antibiotic	3.27±1.39	0.42
B.Comorbidities	1	Diabetes	4.73±1.03	0.22
	2	Chronic Obstructive Pulmonary Disease (COPD)	4.93±0.26	0.05
	3	Pulmonary emphysema	4.87±0.52	0.11
	4	Pulmonary tuberculosis (TB)	4.87±0.52	0.11
	5	Interstitial lung disease	4.67±0.90	0.19
	6	Pulmonary fibrosis	4.67±0.90	0.19
	7	Coronary disease	4.73±0.46	0.10
	8	Hypertension	4.60±0.74	0.16

	9	Cerebral infarction	3.87±1.51	0.39
	10	Bronchiectasia	4.80±0.41	0.09
	11	Bullae of lung	4.33±0.49	0.11
	12	Asthma	4.60±0.83	0.18
	13	Heart failure	4.80±0.41	0.09
	14	Arhythmia	4.67±0.49	0.10
	15	Chronic renal insufficiency	4.60±0.51	0.11
	16	Cerebral hemorrhage	3.87±1.36	0.35
C.Preoperative Complete Blood Count (CBC) Indicators	1	Hemoglobin (g/L)	4.40±0.83	0.19
	2	Neutrophilic granulocyte percentage (%)	4.20±0.94	0.22
	3	Lymphocyte percentage (%)	3.80±1.15	0.30
	4	Monocyte percentage (%)	4.07±0.59	0.15
	5	Percentage of eosinophilic cells (%)	3.13±1.25	0.40
	6	Percentage of basophilic cells (%)	3.20±1.21	0.38
	7	Absolute neutrophil values (10^9/L)	4.53±0.52	0.11
	8	Absolute lymphocytes values (10^9/L)	3.40±1.24	0.37
	9	Absolute Monocyte value (10^9/L)	3.40±1.24	0.37
	10	Absolute oncocytic values (10^9/L)	2.87±1.19	0.41
	11	Absolute basophile values (10^9/L)	2.80±1.21	0.43
	12	Reference value of total white blood cell count (10^9/L)	4.33±0.98	0.23
	13	Total erythrocyte count (10^12/L)	3.67±1.54	0.42
	14	Hematocrit (L/L)	3.07±1.53	0.50
	15	Mean corpuscular volume (fL)	2.67±1.18	0.44
	16	Mean corpuscular hemoglobin (pg)	3.53±1.60	0.45
	17	Mean corpuscular hemoglobin Concentration (MCHC, g/L)	2.53±0.99	0.39
	18	Red cell distribution width-coefficient of variation (RDW-CV, %)	4.07±0.70	0.17
	19	Total number of platelets (10^9/L)	3.67±1.45	0.39
	20	Platelet distribution width (PDW, %)	2.80±1.15	0.41
	21	Mean platelet volume (MPV, fL)	2.67±1.11	0.42
	22	Large platelet ratio (%)	2.73±1.16	0.43
	23	Plateletcrit (%)	4.27±0.59	0.14
D. Preoperative Coagulation function	1	international normalized ratio (INR)	3.40±1.45	0.43
	2	Prothrombin time (s)	3.67±1.45	0.39
	3	Prothrombin time activity level (%)	3.40±1.45	0.43
	4	Activated partial thromboplastin time	3.13±1.51	0.48
	5	Contractinogen (g/L)	4.47±0.64	0.14
	6	Thrombin time (s)	4.47±0.74	0.17
	7	Prothrombin time ratio	3.00±1.51	0.50
E. Preoperative Metabolic panel	1	Albumin (g/L)	4.87±0.35	0.07
	2	A/G	3.67±1.40	0.38
	3	Chlorinum (Cl-, mmol/L)	2.93±1.28	0.44
	4	Calcium (Ca^{2+}, mmol/L)	2.87±1.25	0.43

	5	Magnesium (Mg^{2+}, mmol/L)	2.60±1.12	0.43
	6	Alanine aminotransferase (ALT, U/L)	3.27±1.53	0.47
	7	Aspartate aminotransferase (AST,U/L)	3.33±1.45	0.43
	8	Aspartate transaminase/alanine transaminase	3.20±1.27	0.40
	9	Glutamyl-converting enzyme (U/L)	2.80±0.94	0.34
	10	Alkaline phosphatase (ALP, U/L)	2.80±0.94	0.34
	11	Cholinesterase (Che, kU/L)	2.80±0.94	0.34
	12	Total bilirubin(TBil, μmol/L)	3.40±1.45	0.43
	13	Direct bilirubin (DBil, μmol/L)	3.27±1.39	0.42
	14	Indirect bilirubin (IBil, μmol/L)	3.27±1.39	0.42
	15	Total bile acid (TBA, μmol/L)	2.87±1.06	0.37
	16	Total protein (TP, g/L)	4.47±0.74	0.17
	17	K^+ (mmol/L)	3.80±1.42	0.37
	18	Na^+ (mmol/L)	3.80±1.42	0.37
	19	Globulin (GLB, g/L)	3.13±1.30	0.42
	20	Preglobulin (mg/L)	3.33±1.23	0.37
	21	Urea (mmol/L)	3.73±1.53	0.41
	22	Creatinine (μmol/L)	3.67±1.50	0.41
	23	Estimated glomerular filtration rate (eGFR, mL/min/1.73m^2)	3.47±1.46	0.42
	24	Triglyceride (TG, mmol/L)	3.27±1.28	0.39
	25	Total cholesterol ((Tc, mmol/L)	3.27±1.28	0.39
	26	High density lipoprotein cholesterol (HDL-c, mmol/L)	3.20±1.21	0.38
	27	Low density lipoprotein cholesterin (LDL-c, mmol/L)	3.20±1.21	0.38
	28	Apolipoprotein A1 (g/L)	2.47±0.92	0.37
	29	Apolipoprotein B (g/L)	2.47±0.92	0.37
	30	Creatine kinase (CK, U/L)	3.00±1.25	0.42
	31	Creatine Kinase Isoenzyme (CK-MB, U/L)	2.80±1.27	0.45
	32	Lactic dehydrogenase (LDH, U/L)	2.73±1.28	0.47
	33	A-hydroxybutyrate dehydrogenase (U/L)	2.60±1.24	0.48
	34	Audit value	2.67±1.23	0.46
F.Preoperative two-marker panel for myocardial infarction	1	Myoglobin (ng/ml)	2.87±1.30	0.45
	2	High-Sensitivity cardiac troponin (ng/L)	3.40±1.50	0.44
G.Blood Infectious diseases	1	Human immunodeficiency virus (HIV)	2.73±1.71	0.63
	2	Syphilis	2.40±1.40	0.59
	3	HBV Hepatitis B virus antigen	2.67±1.68	0.63
	4	HCV Hepatitis C virus antigen	2.27±1.49	0.65
H. Preoperative Pulmonary function	1	Diffusing capacity of the lungs for carbon monoxide (DLCO, mmol/min/kPa)	4.47±1.06	0.24

	2	Forced expiratory volume in 1 second/Forced vital capacity predicted value (FEV1/ FVC, %)	4.87±0.35	0.07
	3	FEV1 (L)	5.00±0.00	0.00
	4	Peak Expiratory Flow (PEF, L/s)	4.53±0.74	0.16
	5	Maximum Voluntary Ventilation (MVV, L/min)	5.00±0.00	0.00
	6	Single lung ventilation time	3.93±1.58	0.40
	7	FVC (L)	4.53±0.83	0.18
	8	Maximum inspiratory pressure (MIP)	3.67±1.40	0.38
	9	Maximum expiratory pressure (MEP)	3.67±1.40	0.38
	1	Preoperatively airway colonize bacteria	3.80±1.52	0.40
	2	Preoperative American Society of Anesthesiologists classification (ASA)	4.67±0.62	0.13
	3	Qualitative results of high resolution CT scan (HRCT)	3.47±1.51	0.43
	4	Incision VAS score (4-10)	4.07±0.96	0.24
	5	Eastern Oncology Consortium Physical Status Score>1 (ECOG)	4.27±1.10	0.26
	6	Fatigue/ weakness	3.80±1.42	0.37
	7	Acute Physiology and Chronic Health Evaluation II score≥10 (APACHE II)	3.60±1.45	0.40
I. Other indicators	8	Human leukocyte antigen in peripheral blood (CD14/HLA-DR)	3.27±1.28	0.39
	9	Six-Minute walk test (6-WMT)	4.40±0.74	0.17
	10	Erythrocyte sedimentation rate (ESR, mm/h)	3.40±1.45	0.43
	11	Glycosylated hemoglobin (%)	3.33±1.45	0.43
	12	Arterial blood gas analysis (ABG)	3.87±1.64	0.42
	13	Electrocardiography (ECG)	3.20±1.37	0.43
	14	Preoperative cardiac function	3.40±1.60	0.47
	15	Preoperative vital signs	3.40±1.77	0.52
	16	Preoperative PaO2	4.47±0.64	0.14
	17	Prognostic Nutritional Index (PNI)	4.53±0.64	0.14
	18	Preoperative peak oxygen pulse percentage	4.27±0.80	0.19
	1	Histopathologic type	4.47±0.83	0.19
	2	TNM stage	4.60±0.74	0.16
	3	Surgical method (thoracotomy/thoracoscope)	4.73±1.03	0.22
	4	Peroperative bleeding	4.73±1.03	0.22
J. Surgical indicators	5	Type of operation	5.00±0.00	0.00
	6	Operation duration	4.73±1.03	0.22
	7	Enlargement operation	4.53±1.13	0.25
	8	Lymph node dissection	4.47±1.06	0.24
	9	Intraoperative infusion volume	4.60±0.63	0.14
	10	Duration of anesthesia	4.93±0.26	0.05

	11	Intraoperative volume per minute	3.40±1.77	0.52
	12	Time to extubation	3.40±1.77	0.52
	13	Anaesthetic type	3.40±1.77	0.52
	14	Intraoperative tidal volume	3.40±1.77	0.52
	15	Conversion thoracotomy	4.47±1.13	0.25
	16	Intraoperative blood transfusion volume	3.53±1.81	0.51
	17	Operative site	3.33±1.68	0.50
	18	Anesthesia method	2.80±1.47	0.53
	19	Intubation method	2.60±1.35	0.52
	20	Intraoperative rescue	2.93±1.71	0.58
	21	Cause of rescue	2.93±1.71	0.58
	22	Additional intraoperative use of antibiotics	3.40±1.64	0.48
K. Postoperative indicators	1	Days of antibiotic use in perioperative period	4.40±1.06	0.24
	2	First blood gas after surgery	2.93±1.53	0.52
	3	If using postoperative analgesia	3.13±0.83	0.27
	4	Analgesia	2.80±0.86	0.31
	5	Analgesic duration	4.00±1.41	0.35
	6	Time of the first ambulation	4.67±0.62	0.13
	7	Aerosol inhalation	4.53±0.64	0.14
	8	Nebulized inhalant drugs	3.20±0.94	0.29
	9	Duration of atomizing inhalation	4.20±0.78	0.18
	10	Duration of oxygen therapy	3.27±1.49	0.45
	11	The flow of oxygen therapy	3.33±1.40	0.42
	12	Length of ICU stay	3.00±1.51	0.50
	13	Postoperative use of antibiotics	3.33±1.50	0.45
	14	Antibiotic type	3.40±1.55	0.46
	15	If transfer to ICU after surgery?	3.27±1.49	0.45
	16	Duration of closed thoracic drainage	4.67±0.62	0.13
	17	Thoracic drainage volume on the first day after surgery	4.40±0.83	0.19
	18	Negative pressure Chest drainage volume	3.20±1.32	0.41
	19	Thoracic closed drainage extubation time	3.33±1.45	0.43
	20	Thoracic drainage volume	4.53±0.64	0.14
	21	Reintubate thoracic drainage	4.33±0.62	0.14
	22	Unplanned chest tube extubation	2.87±1.25	0.43
L. pulmonary rehabilitation	1	Abdominal breath training	4.60±0.74	0.16
	2	Pursed-lip breathing training	4.60±0.74	0.16
	3	Blow the balloon traning	4.60±0.74	0.16
	4	Cough training	4.60±0.74	0.16
	5	Stair climbing training	3.13±1.46	0.47

Supplementary Table 3 Variables for Risk Factors of PPCs After Lung Cancer Surgery: Results of the Second Round of Delphi Consultation

Dimension	Indicator	$\bar{X}\pm S$	Coefficient of Variation
A. Demographics Information	A1. Age	4.80±0.41	0.09
	A2. Sex	4.67±0.49	0.10
	A3. Smoking History	5.00±0.00	0.00
	A4. Years of Smoking	4.80±0.41	0.09
	A5. Body Mass Index (BMI)	5.00±0.00	0.00
	A6. Postoperative Hospitalization Duration	5.00±0.00	0.00
	A7. History of Chemotherapy	4.73±0.59	0.13
	A8. History of Radiotherapy	4.93±0.26	0.05
	A9. History of Thoracic Surgery	4.60±0.63	0.14
B. Comorbidity	B1. Diabetes	5.00±0.00	0.00
	B2. Chronic Obstructive Pulmonary Disease (COPD)	5.00±0.00	0.00
	B3. Emphysema	4.73±0.46	0.10
	B4. Tuberculosis	5.00±0.00	0.00
	B5. Pulmonary Fibrosis	4.87±0.35	0.07
	B6. Coronary Artery Disease	4.47±0.52	0.12
	B7. Hypertension	4.47±0.52	0.12
	B8. Bronchiectasis	4.60±0.51	0.11
	B9. Bullous Lung Disease	4.73±0.46	0.10
	B10. Asthma	5.00±0.00	0.00
	B11. Arrhythmia	4.53±0.52	0.11
	B12. Chronic Renal Insufficiency	4.60±0.51	0.11
	B13. Presence of Other Malignant Tumors	4.93±0.26	0.05
	B14. Charson comorbidity index (CCI)	4.87±0.35	0.07
C. Complete Blood Count (CBC) Indicators	C1. Hemoglobin	4.80±0.41	0.09
	C2. Neutrophil Percentage	4.87±0.35	0.07
	C3. Absolute Neutrophil Count	4.80±0.41	0.09
	C4. Monocyte Percentage	4.60±0.51	0.11
	C5. White Blood Cell Count	4.67±0.49	0.10
	C6. Red cell distribution width-coefficient of variation (RDW-CV, %)	4.93±0.26	0.05
	C7. Plateletocrit (PCT, %)	4.73±0.46	0.10
D. Coagulation Test Indicators	D1. Fibrinogen (FIB, g/L)	5.00±0.00	0.00
	D2. Thrombin Time (TT, s)	5.00±0.00	0.00
E. Metabolic panel indicators	E1. Albumin (ALB, g/L)	5.00±0.00	0.00
	E2. Total Protein (TP, g/L)	4.87±0.35	0.07
	E3. A/G Ratio	4.73±0.59	0.13
	E4. Blood Glucose (BG, mmol/L)	5.00±0.00	0.00
F. Pulmonary Function Indicators	F1. Diffusing Capacity of the Lungs for Carbon Monoxide Predicted Value (DLCO, mmol/min/kPa)	4.93±0.26	0.05
	F2. DLCO Measured Value (mmol/min/kPa)	4.87±0.35	0.07
	F3. DLCO Measured/Predicted Percentage (mmol/min/kPa)	4.93±0.26	0.05
	F4. Forced Expiratory Volume in 1 second/Forced Vital Capacity Predicted Value (FEV1/FVC%)	4.93±0.26	0.05
	F5. FEV1/ FVC Measured Value (%)	5.00±0.00	0.00
	F6. FEV1/ FVC Measured/Predicted Percentage (%)	4.93±0.26	0.05
	F7. Forced Expiratory Volume in 1 second Predicted Value (FEV1, L)	4.93±0.26	0.05
	F8. FEV1 Measured Value (L)	5.00±0.00	0.00

	F9. FEV1 Measured/Predicted Percentage (L)	5.00±0.00	0.00
	F10. Forced Vital Capacity Predicted Value (FVC, L)	4.80±0.41	0.09
	F11. FVC Measured Value (L)	4.80±0.41	0.09
	F12.FVC Measured/Predicted Percentage (L)	4.73±0.46	0.10
	F13. Predicted Peak Expiratory Flow Predicted Value (L/s)	4.80±0.41	0.09
	F14. PEF Measured Value (L/s)	4.93±0.26	0.05
	F15. PEF Measured/Predicted Percentage (L/s)	4.73±0.59	0.13
	F16. Maximum Voluntary Ventilation Predicted Value (MVV, L/min)	4.80±0.41	0.09
	F17. MVV Measured Value (L/min)	4.80±0.41	0.09
	F18. MVV Measured/Predicted Percentage (L/min)	4.73±0.46	0.10
G. Other Preoperative Indicators	G. Preoperative American Society of Anesthesiologists classification (ASA)	5.00±0.00	0.00
H. Intraoperative indicators	H1. Histopathology Type	4.87±0.35	0.07
	H2. Tumor Staging	5.00±0.00	0.00
	H3. Conversion from VATS to thoracotomy	4.93±0.26	0.05
	H4. Auxiliary Small Incision	4.93±0.26	0.05
	H5. Type of Surgery	5.00±0.00	0.00
	H6. Pleural Adhesiolysis	4.93±0.26	0.05
	H7. Intercostal Nerve Blockage	4.80±0.41	0.09
	H8. Systematic lymph node dissection	5.00±0.00	0.00
	H9. Surgery Duration (min)	4.87±0.35	0.07
	H10. Intraoperative Blood Loss (ml)	5.00±0.00	0.00
	H11. Intraoperative Blood Transfusion	5.00±0.00	0.00
	H12. Intraoperative Fluid Infusion (ml)	4.67±0.49	0.10

Supplementary Table 4 Summary of Missing Data in model development dataset

Variable item	Number	Missing data	Missing percent
Body Mass Index (BMI)	777	106.00	12.00
White Blood Cell Count (WBC) (10^9/L)	882	1.00	0.10
Neutrophil Percentage (%)	881	2.00	0.20
Monocyte Percentage (%)	882	1.00	0.10
Hemoglobin (Hb) (g/L)	882	1.00	0.10
Red Cell Distribution Width-coefficient of variation (RDW-CV) (%)	882	1.00	0.10
Plateletcrit (PCT) (%)	882	1.00	0.10
Fibrinogen (FIB) (g/L)	882	1.00	0.10
Thrombin Time (TT) (s)	845	38.00	4.30
Total Protein (TP) (g/L)	880	3.00	0.30
Albumin (ALB) (g/L)	880	3.00	0.30
Albumin/Globulin Ratio (A/G)	874	9.00	1.00
Blood Glucose (BG) (mmol/L)	874	9.00	1.00
DLCO Predicted Value (mmol/min/kPa)	874	9.00	1.00
DLCO Measured Value (mmol/min/kPa)	831	52.00	5.90
DLCO Measured/Predicted Percentage (mmol/min/kPa)	750	133.00	15.10
FEV1/FVC Predicted Value (%)	749	134.00	15.20
FEV1/FVC Measured Value (%)	749	134.00	15.20
FEV1/FVC Measured/Predicted Percentage (%)	749	134.00	15.20

FEV1 Predicted Value (L)	750	133.00	15.10
FEV1 Measured Value (L)	751	132.00	14.90
FEV1 Measured/Predicted Percentage (L)	749	134.00	15.20
FVC Predicted Value (L)	750	133.00	15.10
FVC Measured Value (L)	750	133.00	15.10
FVC Measured/Predicted Percentage (L)	749	134.00	15.20
PEF Predicted Value (L/s)	750	133.00	15.10
PEF Measured Value (L/s)	750	133.00	15.10
PEF Measured/Predicted Percentage (L/s)	749	134.00	15.20
MVV Predicted Value (L/min)	749	134.00	15.20
MVV Measured Value (L/min)	748	135.00	15.30
MVV Measured/Predicted Percentage (L/min)	749	134.00	15.20
ASA Score	748	135.00	15.30
Tumor Stage	748	135.00	15.30
Intraoperative Fluid Volume (ml)	731	152.00	17.20

Notes: ASA score: American Society of Anesthesiologists classification; BMI: Body Mass Index; CCI: Charson comorbidity index (CCI); A/G ratio: Albumin-to-globulin ratio; DLCO: Diffusing Capacity of the Lung for Carbon Monoxide; FEV1/FVC%: The percentage of forced expiratory volume in the first second to forced vital capacity; FEV1: Forced Expiratory Volume in the first second; PEF: Peak Expiratory Flow; MVV: Maximum Voluntary Ventilation; FVC%: Forced Vital Capacity percentage.

Supplementary Table 5 First Round of Collinearity Analysis of Risk Factors

	Variable	Unstandardized Coefficients	Std. Error	Standardized Coefficients	*t*	P	Collinearity Statistics	
		B		Beta			Tolerance	VIF
	(Constant)	-1.144	1.066		-1.073	0.283		
1	Sex	-0.001	0.084	-0.001	-0.013	0.99	0.104	9.621
2	Smoking	-0.048	0.072	-0.039	-0.671	0.503	0.225	4.448
3	Preoperative Chemotherapy	-0.086	0.108	-0.032	-0.803	0.422	0.487	2.055
4	Neoadjuvant chemotherapy	0	0.128	0	0.002	0.999	0.505	1.982
5	Hypertension	-0.007	0.036	-0.006	-0.202	0.84	0.797	1.254
6	Diabetes	-0.005	0.062	-0.002	-0.078	0.938	0.787	1.271
7	Coronary Artery Disease	0.202	0.111	0.053	1.817	0.07	0.892	1.121
8	ASA score	-0.016	0.033	-0.014	-0.5	0.617	0.9	1.112
9	Pathological	-0.03	0.085	-0.012	-0.353	0.724	0.633	1.579
10	Cancer Stage	0.072	0.026	0.081	2.727	0.007	0.861	1.162
11	Conversion to thoracotomy	-0.044	0.135	-0.01	-0.324	0.746	0.842	1.187
12	Auxiliary Small Incision	0.257	0.073	0.123	3.505	0	0.611	1.636
13	Type of Surgery	0.019	0.008	0.069	2.423	0.016	0.932	1.073
14	Type of Lymph Node Dissection	0.011	0.012	0.027	0.949	0.343	0.925	1.081
15	Thoracic Adhesionolysis	0.161	0.05	0.097	3.236	0.001	0.846	1.181
16	Intraoperative blood transfusion volume	-0.004	0.076	-0.002	-0.056	0.956	0.742	1.349
17	Age	0.009	0.004	0.219	2.631	0.009	0.109	9.137

18	BMI	0.008	0.005	0.049	1.597	0.111	0.805	1.243
19	Year of smoking	0.004	0.002	0.103	1.778	0.076	0.224	4.466
20	CCI score	0.016	0.015	0.049	1.07	0.285	0.36	2.779
21	Monocyte percentage (%)	0.171	0.691	0.008	0.247	0.805	0.787	1.27
22	RDW-CV (%)	0.016	0.011	0.044	1.467	0.143	0.825	1.211
23	Plateletcrit (%)	-0.015	0.014	-0.03	-1.077	0.282	0.947	1.056
24	Fibrinogen (g/L)	0.005	0.015	0.009	0.316	0.752	0.854	1.171
25	Albumin (g/L)	0.008	0.005	0.054	1.666	0.096	0.708	1.412
26	A/G ratio (%)	-0.029	0.052	-0.018	-0.55	0.583	0.742	1.347
27	Blood glucose (mmol/L)	0.007	0.011	0.02	0.663	0.508	0.798	1.253
28	DLCO predicted (mmol/min/kPa)	-0.312	0.13	-0.716	-2.409	0.016	0.009	116.891
29	DLCO measured (mmol/min/kPa)	-0.011	0.016	-0.033	-0.687	0.492	0.319	3.131
30	FEV1/FVC predicted (%)	0	0.012	-0.006	-0.034	0.973	0.022	45.548
31	FEV1 predicted (L)	0.383	0.471	0.447	0.813	0.416	0.003	399.986
32	FEV1 measured (L)	-0.154	0.286	-0.196	-0.538	0.591	0.006	174.93
33	FEV1 actual/predicted (%)	0	0.008	0.008	0.029	0.977	0.01	103.809
34	FEV1/FVC actual/predicted (%)	0.007	0.007	0.133	0.967	0.334	0.04	24.898
35	FVC predicted (L)	0.686	0.349	0.998	1.965	0.05	0.003	341.37
36	FVC measured (L)	-0.232	0.231	-0.348	-1.005	0.315	0.006	159.124
37	FVC actual/predicted (%)	0.008	0.008	0.236	0.933	0.351	0.012	84.618
38	PEF measured (L/s)	0.003	0.067	0.011	0.049	0.961	0.014	71.392
39	PEF actual/predicted (%)	0	0.004	-0.004	-0.025	0.98	0.028	35.988
40	MVV predicted (L/min)	-0.01	0.007	-0.337	-1.41	0.159	0.013	75.567
41	MVV measured (L/min)	0.003	0.003	0.133	0.776	0.438	0.026	39.017
42	MVV actual/predicted (%)	-0.003	0.003	-0.1	-0.823	0.411	0.051	19.663
43	Surgery duration (min)	0.002	0	0.149	4.33	0	0.641	1.559
44	Intraoperative blood loss (ml)	0	0	0.03	0.836	0.404	0.603	1.66
45	Intraoperative fluid volume (ml)	8.964E-5	0	0.054	1.703	0.089	0.763	1.31

Note:; ASA score: American Society of Anesthesiologists classification; BMI: Body Mass Index; CCI: Charson comorbidity index (CCI); A/G ratio: Albumin-to-globulin ratio; DLCO: Diffusing Capacity of the Lung for Carbon Monoxide; FEV1/FVC%: The percentage of forced expiratory volume in the first second to forced vital capacity; FEV1: Forced Expiratory Volume in the first second; PEF: Peak Expiratory Flow; MVV: Maximum Voluntary Ventilation; FVC%: Forced Vital Capacity percentage, RDWCV:Red Blood Cell Distribution Width.

Supplementary Table 6 Second Round of Collinearity Analysis of Risk Factors

	Variable	Unstandardized Coefficients	Std. Error	Standardized Coefficients	t	P	Collinearity Statistics	
		B		Beta			Tolerance	VIF
	(Constant)	-1.325	0.345		-3.844	0		
1	Sex	0.084	0.039	0.086	2.126	0.034	0.472	2.118
2	Smoking	-0.054	0.071	-0.044	-0.76	0.447	0.231	4.321
3	Preoperative Chemotherapy	-0.126	0.107	-0.046	-1.173	0.241	0.495	2.019
4	Neoadjuvant chemotherapy	0.055	0.127	0.017	0.429	0.668	0.515	1.943
5	Hypertension	0.002	0.036	0.002	0.06	0.952	0.807	1.239
6	Diabetes	-0.005	0.062	-0.002	-0.08	0.936	0.798	1.253
7	Coronary Artery Disease	0.172	0.111	0.045	1.552	0.121	0.914	1.095
8	ASA score	-0.021	0.032	-0.018	-0.642	0.521	0.94	1.064
9	Pathological	-0.033	0.084	-0.014	-0.395	0.693	0.653	1.531
10	Tumor Stage	0.076	0.026	0.086	2.926	0.004	0.887	1.128
11	Conversion to Thoracotomy	-0.019	0.134	-0.004	-0.145	0.885	0.865	1.155
12	Auxiliary Small Incision	0.281	0.073	0.135	3.845	0	0.623	1.604
13	Type of Surgery	0.02	0.008	0.071	2.482	0.013	0.946	1.057
14	Type of Lymph Node Dissection	0.011	0.012	0.027	0.939	0.348	0.931	1.074
15	Thoracic Adhesionolysis	0.16	0.05	0.096	3.236	0.001	0.863	1.159
16	Intraoperative blood transfusion volume	-0.003	0.076	-0.001	-0.035	0.972	0.757	1.322
17	Age	0.012	0.002	0.285	5.795	0	0.315	3.176
18	BMI	0.007	0.005	0.047	1.549	0.122	0.842	1.187
19	Year of smoking	0.005	0.002	0.112	1.97	0.049	0.236	4.231
20	CCI score	0.021	0.015	0.065	1.435	0.152	0.371	2.699
21	Monocyte percentage (%)	0.292	0.687	0.013	0.424	0.671	0.806	1.241
22	RDWCV (%)	0.013	0.011	0.037	1.236	0.217	0.855	1.17
23	Plateletcrit (%)	-0.014	0.014	-0.03	-1.052	0.293	0.953	1.049
24	Fibrinogen (g/L)	0.006	0.015	0.012	0.409	0.683	0.882	1.134
25	Albumin (g/L)	0.009	0.005	0.063	1.973	0.049	0.742	1.347
26	A/G ratio (%)	-0.033	0.052	-0.02	-0.623	0.534	0.755	1.324
27	Blood glucose (mmol/L)	0.005	0.011	0.014	0.445	0.656	0.831	1.204
28	Surgery duration (min)	0.001	0	0.143	4.202	0	0.657	1.523
29	Intraoperative blood loss (ml)	0	0	0.028	0.786	0.432	0.61	1.64
30	Intraoperative fluid volume (ml)	0	0	0.062	2.007	0.045	0.811	1.233
31	DLCO measured (mmol/min/kPa)	-0.023	0.013	-0.071	-1.76	0.079	0.474	2.111

Note: ASA score: American Society of Anesthesiologists classification; BMI: Body Mass Index; CCI: Charson comorbidity index (CCI); A/G ratio: Albumin-to-globulin ratio; DLCO: Diffusing Capacity of the Lung for Carbon Monoxide; RDWCV:Red Blood Cell Distribution Width.

Supplementary Table 7 The feature rank in 9 Machine learning algorithms

Feature Rank	Features	Boruta	DT	GBDT	GNB	Light GBM	Logit	PLSDA	RF	XGBoost	Comprehensive Ranking
1	Age	1	2	2	1	3	1	1	1	11	2.556
2	Surgery duration	1	3	3	6	1	1	3	3	9	3.333
3	CCI score	1	1	1	25	14	9	2	2	2	6.333
4	Tumor stage	1	19	4	4	16	1	5	4	4	6.444
5	DLCO measured	1	4	5	21	2	9	11	5	16	8.222
6	Intraoperative fluid volume	1	8	10	17	9	9	14	6	12	9.556
7	RDW-CV	1	13	8	13	7	1	16	9	22	10.000
8	BMI	1	7	7	9	4	9	22	8	25	10.222
9	Year of smoking	2	14	12	2	20	9	6	21	8	10.444
10	Fibrinogen	1	10	9	10	5	9	20	7	23	10.444
11	Sex	1	11	22	15	18	1	10	15	15	12.000
12	Albumin	6	9	11	11	8	9	26	10	21	12.333
13	Pleural adhesion release	4	15	15	24	23	1	9	17	3	12.333
14	Blood glucose	1	12	6	23	6	9	24	11	20	12.444
15	Surgery type	4	17	13	19	13	1	27	13	13	13.333
16	Hypertension	7	21	20	3	19	9	13	22	7	13.444
17	Auxiliary	4	24	23	18	22	1	4	20	5	13.444
18	Intraoperative blood loss	9	18	18	16	17	9	12	16	14	14.333
19	Smoking	9	26	17	14	24	9	7	19	6	14.556
20	A/G ratio	7	6	16	28	11	9	28	14	17	15.111
21	Plateletcrit	11	5	14	26	10	9	31	12	19	15.222
22	Diabetes	17	20	26	7	26	9	21	26	10	18.000
23	ASA score	14	22	21	8	21	9	25	24	18	18.000
24	Monocyte percentage	12	25	19	30	12	9	19	18	26	18.889
25	Lymph node dissection	13	16	24	20	15	9	30	23	24	19.333
26	Intraoperative blood transfusion	16	23	27	5	28	9	18	28	27	20.111
27	Pathological typel	15	30	30	12	27	9	8	25	29	20.556
28	Coronary heart disease	20	29	25	27	30	9	23	27	1	21.222
29	Preoperative chemotherapy	18	27	29	22	25	9	15	29	28	22.444
30	Preoperative Neoadjuvant chemotherapy	19	28	28	29	29	9	17	30	30	24.333
31	Conversion to thoracotomy	21	31	31	31	31	9	29	31	31	27.222

Note: ASA score: American Society of Anesthesiologists classification; BMI: Body Mass Index; CCI: Charson comorbidity index (CCI); A/G ratio: Albumin-to-globulin ratio; DLCO: Diffusing Capacity of the Lung for Carbon Monoxide; : Logit: Logistic Regression; LightGBM: Light Gradient Boosting Machine; RF: Random Forest; GNB: Gaussian Naive Bayesian; KNN: k-Nearest Neighbor; DT: Decision Tree; XGBoost: Extreme Gradient Boosting, GBDT, Gradient Boosting Decision Tree; PLSDA, Partial Least Squares Discriminant Analysis.RDWCV:Red Blood Cell Distribution Width.

Supplementary Table 8 The comparison of characteristics between the model development and prospective validation dataset

Variable	Model development dataset (N=883)	Prospective validation dataset (N=308)	χ^2/Z	P
PPCs				
No	570(64.6)	217(70.7)	3.824	0.059
Yes	313(35.4)	91(29.5)		
Sex, n(%)				
Female	540(61.2)	177(57.5)	1.296[a]	0.255
Male	343(38.8)	131(42.5)		
Smoking history, n(%)				
No	722(81.8)	235(76.3)	4.325[a]	0.038
Yes	161(18.2)	73(23.7)		
Preoperative chemotherapy, n(%)				
No	855(96.8)	292(94.8)	2.629[a]	0.105
Yes	28(3.2)	16(5.2)		
Preoperative Neoadjuvant chemotherapy				
No	864(97.8)	298(96.8)	1.153[a]	0.283
Yes	19(2.2)	10(3.2)		
History of thoracic surgery, n(%)				
No	882(99.9)	298(96.8)	21.198[a]	**<0.001**
Yes	1(0.1)	10(2.8)		
Hypertension, n(%)				
No	692(78.4)	236(76.6)	0.404[a]	0.525
Yes	191(21.6)	72(23.4)		
Diabetes, n(%)				
No	830(94.0)	284(92.2)	1.210[a]	0.271
Yes	53(6.0)	24(7.8)		
Coronary heart disease, n(%)				
No	869(98.4)	298(96.8)	3.192[a]	0.074
Yes	14(1.6)	10(3.2)		
Arrhythmia, n(%)				
No	879(99.5)	286(92.9)	47.858[a]	<.001
Yes	4(0.5)	22(7.1)		
Chronic renal insufficiency, n(%)				
No	881(99.8)	306(99.4)	0.284[a]	0.594
Yes	2(0.2)	2(0.6)		
COPD, n(%)				
No	876(99.2)	294(95.5)	18.565[a]	**<0.001**
Yes	7(0.8)	14(4.5)		
Bronchiectasis, n(%)				
No	879(99.5)	305(99.0)	0.357[a]	0.55
Yes	4(0.5)	3(1.0)		
Pulmonary bullae, n(%)				
No	877(99.3)	296(96.1)	13.785[a]	**<0.001**
Yes	6(0.7)	12(3.9)		

Asthma, n(%)				
No	879(99.5)	306(99.4)	0.000[a]	1
Yes	4(0.5)	2(0.6)		
Pulmonary tuberculosis, n(%)				
No	858(97.2)	298(96.8)	0.138[a]	0.71
Yes	25(2.8)	10(3.2)		
Pulmonary fibrosis, n(%)				
No	880(99.7)	305(99.0)	0.786[a]	0.375
Yes	3(0.3)	3(1.0)		
Emphysema, n(%)				
No	878(99.4)	289(93.8)	36.301[a]	**<0.001**
Yes	5(0.6)	19(6.2)		
With other tumors, n(%)				
No	792(89.7)	281(91.2)	0.606[a]	0.436
Yes	91(10.3)	27(8.8)		
ASA score, n(%)				
Grade 1	175(19.8)	2(0.6)		
Grade 2	696(78.8)	283(91.9)	90.196[a]	**＜0.001**
Grade 3	12(1.4)	23(7.5)		
Pathological type, n(%)				
Squamous cell carcinoma	35(4.0)	26(8.4)	9.422[a]	**0.002**
Adenocarcinoma	848(96.0)	282(91.6)		
Tumor stage, n(%)				
Stage I	697(78.9)	233(75.6)		
Stage II	155(17.6)	62(20.2)	2.883[a]	0.405
Stage III	24(2.7)	8(2.6)		
Stage IV	7(0.8)	5(1.6)		
Conversion to thoracotomy, n(%)				
No	873(98.9)	306(99.4)	0.160[a]	0.689
Yes	10(1.1)	2(0.6)		
Use of auxiliary small incision, n(%)				
No	834(94.5)	293(95.1)	0.207[a]	0.649
Yes	49(5.5)	15(4.9)		
Surgery type, n(%)				
Sleeve resection	2(0.2)	1(0.3)		
Lobectomy	436(49.4)	130(42.2)		
Segmentectomy	203(23.0)	58(18.8)		
Wedge resection	69(7.8)	38(12.3)	21.145[a]	0.004
Pneumonectomy	6(0.7)	3(1.0)		
Lobectomy+ wedge resection	110(12.5)	61（19.8）		
Segmentectomy + wedge resection	30(3.4)	13(4.2)		
Lobectomy + segmentectomy	27(3.1)	4(1.3)		
Lymph node dissection, n(%)				
No	310(35.1)	25(8.1)	233.677[a]	**＜0.001**
Intrathoracic	119(13.5)	166(53.9)		

Systematic	296(33.5)	56(18.2)		
Mediastinal	158(17.9)	61(19.8)		
Pleural adhesion release, n(%)				
No	803(90.9)	272(88.3)	1.794[a]	0.18
Yes	80(9.1)	36(11.7)		
Intercostal nerve block, n(%)				
No	880(99.7)	239(77.6)	196.697[a]	**<0.001**
Yes	3(0.3)	69(22.4)		
Intraoperative blood transfusion, n(%)				
No	846(95.8)	68(22.1)	695.516[a]	**<0.001**
Yes	37(4.2)	240(77.9)		
Age (years)	55.00(48.00, 62.00)	55.00(49.00, 61.00)	-0.052[b]	0.959
BMI	23.50(21.69, 25.54)	23.70(21.56, 25.97)	-1.118[b]	0.264
Year of smoking (years)	0.00(0.00, 0.00)	0.00(0.00, 0.00)	-2.196[b]	**0.028**
Charlson Comorbidity Index (CCI)	2.00(1.00, 3.00)	2.00(2.00, 3.00)	-1.749[b]	0.08
White blood cell count (10^9/L)	5.36(4.53, 6.42)	5.67(4.70, 7.04)	-3.047[b]	**0.002**
Neutrophil percentage (%)	0.57(0.51, 0.63)	0.57(0.51, 0.63)	-1.032[b]	0.302
Monocyte percentage (%)	0.08(0.07, 0.09)	0.08(0.07, 0.09)	-0.057[b]	0.954
Absolute neutrophil count (10^9/L)	3.07(2.37, 3.76)	3.23(2.42, 4.18)	-2.589[b]	**0.01**
Hemoglobin (g/L)	130(121.00, 141.00)	132.00(121.00, 143.00)	-1.201[b]	0.23
Red blood cell distribution width (%)	13.10(12.50, 13.60)	12.90(12.40, 13.60)	-1.850[b]	0.064
Plateletcrit (%)	0.23(0.19, 0.27)	0.23(0.20, 0.26)	-0.575[b]	0.565
Fibrinogen (g/L)	2.72(2.34, 3.26)	2.64(2.31, 3.15)	-1.445[b]	0.148
Thrombin time (s)	17.70(17.10, 18.40)	17.80(17.30, 18.50)	-2.170[b]	0.03
Total protein (g/L)	63.80(60.70, 67.10)	64.40(60.83, 67.80)	-0.887[b]	0.375
Albumin (g/L)	40.40(38.40, 42.60)	39.90(37.93, 41.80)	-2.949[b]	**0.003**
A/G ratio (%)	1.70(1.60, 1.90)	1.68(1.50, 1.88)	-4.351[b]	**<0.001**
Blood glucose (mmol/L)	4.58(4.19, 5.07)	4.81(4.41, 5.39)	-4.726[b]	**<0.001**
DLCO predicted (mmol/min/kPa)	7.70(7.06, 8.42)	7.70(7.00, 8.75)	-1.096[b]	0.273
DLCO measured (mmol/min/kPa)	7.76(6.80, 8.77)	7.80(6.72, 8.87)	-0.453[b]	0.65
DLCO actual/predicted (%)	101.00(93.00, 110.00)	101.00(91.10, 109.00)	-0.773[b]	0.439
FEV1/FVC predicted (%)	84.06(83.86, 84.25)	84.02(83.77, 84.25)	-1.459[b]	0.145
FEV1/FVC measured (%)	79.18(75.11, 82.99)	79.16(73.84, 83.74)	-0.391[b]	0.696
FEV1/FVC actual/predicted (%)	94.70(89.60, 98.90)	94.55(87.79, 99.90)	-0.751[b]	0.453
FEV1 predicted (L)	2.42(2.09, 2.83)	2.49(2.07, 2.99)	-1.290[b]	0.197
FEV1 measured (L)	2.40(2.06, 2.81)	2.42(2.05, 2.94)	-0.330[b]	0.742
FEV1 actual/predicted (%)	100.30(91.70, 109.34)	100.65(89.73, 110.50)	-0.115[b]	0.908
FVC predicted (L)	2.90(2.49, 3.51)	2.97(2.48, 3.63)	-1.142[b]	0.253
FVC measured (L)	3.04(2.65, 3.58)	3.14(2.63, 3.62)	-0.948[b]	0.343
FVC actual/predicted (%)	104.68(96.40, 113.20)	103.97(94.70, 114.75)	-0.531[b]	0.595
PEF predicted (L/s)	6.26(5.71, 7.52)	6.42(5.70, 7.71)	-1.152[b]	0.249
PEF measured (L/s)	6.31(5.41, 7.51)	6.51(5.51, 7.63)	-1.384[b]	0.166
PEF actual/predicted (%)	98.97(87.50, 109.90)	100.25(88.65, 112.94)	-1.234[b]	0.217
MVV predicted (L/min)	95.87(87.07, 108.62)	96.19(86.12, 111.93)	-0.908[b]	0.364
MVV measured (L/min)	93.33(78.40, 110.50)	94.90(79.47, 114.03)	-1.171[b]	0.242

MVV actual/predicted (%)	96.80(86.50, 106.50)	98.90(86.65, 112.60)	-1.575[b]	0.115
Surgery duration (min)	135.00(110.00, 165.00)	125.00(102.75, 160.00)	-2.747[b]	**0.006**
Intraoperative blood loss (ml)	50.00(20.00, 50.00)	50.00(20.00, 50.00)	-3.838[b]	<.001
Intraoperative fluid volume (ml)	850.00(750.00, 1100.00)	700.00(600.00, 1100.00)	-4.363[b]	<.001

Note: Continuous variables that do not follow a normal distribution are expressed as M(P25, P75); binary categorical variables are presented as percentages. *a indicates the result of the chi-square test, and *b indicates the result of the non-parametric test. COPD: Chronic Obstructive Pulmonary Disease; ASA score: American Society of Anesthesiologists classification; BMI: Body Mass Index; CCI: Charson comorbidity index (CCI); A/G ratio: Albumin-to-globulin ratio; DLCO: Diffusing Capacity of the Lung for Carbon Monoxide; FEV1/FVC%: The percentage of forced expiratory volume in the first second to forced vital capacity; FEV1: Forced Expiratory Volume in the first second; PEF: Peak Expiratory Flow; MVV: Maximum Voluntary Ventilation; FVC%: Forced Vital Capacity percentage.

Supplementary Table 9 Overall performance for the 12 ML models in training dataset

ML	AUC	Accuracy	F1_score	NPV	PPV	Sensitivity	Specificity
BPNN	1(0.999-1.001)	0.735	0.753	0.774	0.705	0.807	0.663
Boruta	0.735(0.704-0.766)	1.000	1.000	1.000	1.000	1.000	1.000
DT	0.956(0.942-0.97)	0.956	0.956	0.946	0.967	0.945	0.967
GBDT	0.999(0.996-1.001)	0.999	0.999	0.997	1.000	0.997	1.000
GNB	0.78(0.751-0.809)	0.780	0.769	0.756	0.809	0.734	0.827
KNN	0.851(0.826-0.875)	0.851	0.841	0.812	0.900	0.789	0.912
LightGBM	1(0.999-1.001)	1.000	1.000	1.000	1.000	1.000	1.000
Logit	0.779(0.75-0.808)	0.779	0.779	0.779	0.779	0.779	0.779
PLSDA	0.764(0.734-0.793)	0.764	0.735	0.717	0.837	0.656	0.872
RF	0.995(0.99-1)	0.995	0.995	0.993	0.997	0.992	0.997
SVM	0.779(0.75-0.808)	0.779	0.778	0.776	0.782	0.774	0.784
XGBoost	1(0.999-1.001)	1.000	1.000	1.000	1.000	1.000	1.000

Note: BPNN,Back-Propagation Neural Network; DT, Decision Tree; GBDT, Gradient Boosting Decision Tree; GNB, Gaussian Naïve Bayesian;KNN, k-Nearest Neighbor; LightGBM, Light Gradient Boosting Machine; Logit, Logistic Regression; PLSDA, Partial Least Squares Discriminant Analysis; RF,Random Forest; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 10 Ten-fold cross validation results of 12 ML models

ML	AUC	Accuracy	F1_score	NPV	PPV	Sensitivity	Specificity
BPNN	0.784(0.693-0.875)	0.749	0.749	0.758	0.744	0.765	0.738
Boruta	0.763(0.669-0.857)	0.829	0.828	0.831	0.826	0.834	0.824
DT	**0.763(0.669-0.857)**	**0.766**	**0.763**	**0.751**	**0.781**	**0.753**	**0.783**
GBDT	**0.832(0.749-0.915)**	**0.842**	**0.840**	**0.838**	**0.845**	**0.842**	**0.845**
GNB	0.843(0.762-0.924)	0.768	0.750	0.748	0.796	0.720	0.816
KNN	0.811(0.725-0.898)	0.808	0.814	0.848	0.773	0.864	0.752
LightGBM	0.837(0.755-0.919)	0.839	0.837	0.841	0.835	0.844	0.832
Logit	**0.822(0.737-0.907)**	**0.778**	**0.775**	**0.779**	**0.774**	**0.782**	**0.775**
PLSDA	0.777(0.685-0.869)	0.751	0.716	0.711	0.812	0.648	0.851
RF	**0.864(0.787-0.94)**	**0.839**	**0.835**	**0.837**	**0.839**	**0.835**	**0.841**
SVM	**0.811(0.725-0.898)**	**0.771**	**0.769**	**0.775**	**0.766**	**0.779**	**0.766**
XGBoost	0.805(0.717-0.893)	0.828	0.823	0.814	0.838	0.812	0.844

Note: BPNN, Back-Propagation Neural Network; DT, Decision Tree; GBDT, Gradient Boosting Decision Tree; GNB, Gaussian Naïve Bayesian; KNN, k-Nearest Neighbor; LightGBM, Light Gradient Boosting Machine; Logit, Logistic Regression; PLSDA, Partial Least Squares Discriminant Analysis; RF,Random Forest; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 11 The AUC of each-fold cross validation of 12 ML models

Model	Fold_1	Fold_2	Fold_3	Fold_4	Fold_5	Fold_6	Fold_7	Fold_8	Fold_9	Fold_10
Boruta	0.828(0.745-0.912)	0.875(0.802-0.948)	0.778(0.686-0.869)	0.788(0.698-0.878)	0.872(0.799-0.946)	0.823(0.739-0.907)	0.862(0.785-0.938)	0.874(0.801-0.948)	0.804(0.716-0.892)	0.784(0.693-0.875)
BPNN	0.792(0.703-0.881)	0.74(0.644-0.837)	0.786(0.695-0.876)	0.731(0.633-0.828)	0.716(0.616-0.815)	0.764(0.671-0.857)	0.772(0.679-0.864)	0.758(0.663-0.853)	0.695(0.594-0.797)	0.763(0.669-0.857)
DT	0.781(0.691-0.872)	0.799(0.71-0.887)	0.59(0.483-0.698)	0.764(0.67-0.857)	0.787(0.697-0.877)	0.806(0.718-0.893)	0.819(0.733-0.904)	0.784(0.692-0.875)	0.786(0.696-0.877)	0.763(0.669-0.857)
GBDT	0.824(0.74-0.908)	0.849(0.77-0.928)	0.797(0.708-0.885)	0.84(0.759-0.921)	0.871(0.796-0.945)	0.854(0.775-0.932)	0.851(0.772-0.93)	0.858(0.78-0.936)	0.855(0.777-0.933)	0.832(0.749-0.915)
GNB	0.771(0.678-0.863)	0.749(0.654-0.844)	0.778(0.686-0.869)	0.757(0.663-0.851)	0.72(0.621-0.818)	0.785(0.695-0.876)	0.733(0.635-0.83)	0.821(0.736-0.906)	0.724(0.625-0.823)	0.843(0.762-0.924)
KNN	0.819(0.735-0.904)	0.775(0.683-0.867)	0.805(0.717-0.892)	0.809(0.722-0.895)	0.739(0.642-0.835)	0.821(0.736-0.905)	0.784(0.693-0.875)	0.827(0.744-0.911)	0.892(0.823-0.961)	0.811(0.725-0.898)
LightGBM	0.821(0.737-0.906)	0.862(0.786-0.938)	0.808(0.721-0.895)	0.814(0.728-0.9)	0.871(0.796-0.945)	0.837(0.756-0.919)	0.814(0.728-0.9)	0.852(0.774-0.931)	0.863(0.787-0.94)	0.837(0.755-0.919)
Logit	0.772(0.68-0.865)	0.811(0.725-0.898)	0.789(0.699-0.879)	0.751(0.656-0.846)	0.749(0.653-0.844)	0.812(0.726-0.898)	0.772(0.679-0.864)	0.784(0.693-0.875)	0.727(0.628-0.825)	0.822(0.737-0.907)
PLSDA	0.798(0.71-0.887)	0.772(0.679-0.864)	0.706(0.606-0.806)	0.707(0.607-0.807)	0.741(0.645-0.837)	0.715(0.615-0.814)	0.716(0.617-0.816)	0.81(0.723-0.897)	0.755(0.66-0.85)	0.777(0.685-0.869)
RF	0.845(0.765-0.925)	0.786(0.696-0.876)	0.779(0.688-0.871)	0.841(0.761-0.922)	0.886(0.816-0.956)	0.848(0.769-0.928)	0.808(0.721-0.896)	0.884(0.813-0.956)	0.838(0.756-0.92)	0.864(0.787-0.94)
SVM	0.809(0.722-0.895)	0.811(0.725-0.898)	0.778(0.686-0.869)	0.739(0.642-0.835)	0.747(0.651-0.842)	0.812(0.726-0.898)	0.761(0.667-0.855)	0.753(0.658-0.848)	0.704(0.603-0.805)	0.811(0.725-0.898)
XGBoost	0.804(0.717-0.892)	0.849(0.77-0.928)	0.778(0.686-0.869)	0.84(0.759-0.921)	0.847(0.768-0.927)	0.826(0.742-0.909)	0.841(0.759-0.922)	0.868(0.793-0.943)	0.823(0.739-0.908)	0.805(0.717-0.893)

Note: BPNN, Back-Propagation Neural Network; DT, Decision Tree; GBDT, Gradient Boosting Decision Tree; GNB, Gaussian Naïve Bayesian; KNN, k-Nearest Neighbor; LightGBM, Light Gradient Boosting Machine; Logit, Logistic Regression; PLSDA, Partial Least Squares Discriminant Analysis; RF,Random Forest; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 12 Results of Delong test of the 12 ML models

Method 1	Method 2	AUC of Method 1 (95%CI)	AUC of Method 2 (95%CI)	*Z*	*p*
Boruta	BPNN	0.730(0.665-0.796)	0.807(0.749-0.865)	2.151	0.032
Boruta	DT	0.730(0.665-0.796)	0.816(0.759-0.873)	-0.285	0.776
Boruta	GBDT	0.730(0.665-0.796)	0.829(0.774-0.885)	-0.839	0.401
Boruta	GNB	0.730(0.665-0.796)	0.734(0.669-0.799)	1.973	0.049
Boruta	KNN	0.730(0.665-0.796)	0.703(0.636-0.771)	2.540	0.011
Boruta	LightGBM	0.730(0.665-0.796)	0.832(0.777-0.887)	-0.996	0.319
Boruta	Logit	0.730(0.665-0.796)	0.818(0.761-0.875)	-0.349	0.727
Boruta	PLSDA	0.730(0.665-0.796)	0.806(0.748-0.865)	0.033	0.974
Boruta	RF	0.730(0.665-0.796)	0.825(0.769-0.881)	-0.708	0.479
Boruta	SVM	0.730(0.665-0.796)	0.788(0.728-0.848)	0.616	0.538
Boruta	XGBoost	0.730(0.665-0.796)	0.808(0.75-0.866)	-0.022	0.983
BPNN	DT	0.807(0.749-0.865)	0.816(0.759-0.873)	-2.120	0.034
BPNN	GBDT	0.807(0.749-0.865)	0.829(0.774-0.885)	-2.833	0.005
BPNN	GNB	0.807(0.749-0.865)	0.734(0.669-0.799)	-0.122	0.903
BPNN	KNN	0.807(0.749-0.865)	0.703(0.636-0.771)	0.724	0.469
BPNN	LightGBM	0.807(0.749-0.865)	0.832(0.777-0.887)	-2.892	0.004
BPNN	Logit	0.807(0.749-0.865)	0.818(0.761-0.875)	-2.927	0.003
BPNN	PLSDA	0.807(0.749-0.865)	0.806(0.748-0.865)	-2.460	0.014
BPNN	RF	0.807(0.749-0.865)	0.825(0.769-0.881)	-2.984	0.003
BPNN	SVM	0.807(0.749-0.865)	0.788(0.728-0.848)	-1.948	0.051
BPNN	XGBoost	0.807(0.749-0.865)	0.808(0.75-0.866)	-2.122	0.034
DT	**GBDT**	**0.816(0.759-0.873)**	**0.829(0.774-0.885)**	**-0.451**	**0.652**
DT	GNB	0.816(0.759-0.873)	0.734(0.669-0.799)	2.007	0.045
DT	KNN	0.816(0.759-0.873)	0.703(0.636-0.771)	2.620	0.009
DT	LightGBM	0.816(0.759-0.873)	0.832(0.777-0.887)	-0.527	0.598
DT	**Logit**	**0.816(0.759-0.873)**	**0.818(0.761-0.875)**	**-0.053**	**0.958**
DT	PLSDA	0.816(0.759-0.873)	0.806(0.748-0.865)	0.273	0.785
DT	**RF**	**0.816(0.759-0.873)**	**0.825(0.769-0.881)**	**-0.306**	**0.760**
DT	**SVM**	**0.816(0.759-0.873)**	**0.788(0.728-0.848)**	**0.764**	**0.445**
DT	XGBoost	0.816(0.759-0.873)	0.808(0.75-0.866)	0.267	0.790
GBDT	GNB	0.829(0.774-0.885)	0.734(0.669-0.799)	2.771	0.006
GBDT	KNN	0.829(0.774-0.885)	0.703(0.636-0.771)	3.360	0.001
GBDT	LightGBM	0.829(0.774-0.885)	0.832(0.777-0.887)	-0.120	0.905
GBDT	**Logit**	**0.829(0.774-0.885)**	**0.818(0.761-0.875)**	**0.415**	**0.678**
GBDT	PLSDA	0.829(0.774-0.885)	0.806(0.748-0.865)	0.785	0.432
GBDT	**RF**	**0.829(0.774-0.885)**	**0.825(0.769-0.881)**	**0.222**	**0.825**
GBDT	**SVM**	**0.829(0.774-0.885)**	**0.788(0.728-0.848)**	**1.382**	**0.167**
GBDT	XGBoost	0.829(0.774-0.885)	0.808(0.75-0.866)	0.852	0.394
GNB	KNN	0.734(0.669-0.799)	0.703(0.636-0.771)	0.769	0.442
GNB	LightGBM	0.734(0.669-0.799)	0.832(0.777-0.887)	-2.905	0.004
GNB	Logit	0.734(0.669-0.799)	0.818(0.761-0.875)	-2.842	0.004
GNB	PLSDA	0.734(0.669-0.799)	0.806(0.748-0.865)	-2.434	0.015
GNB	RF	0.734(0.669-0.799)	0.825(0.769-0.881)	-2.935	0.003

GNB	SVM	0.734(0.669-0.799)	0.788(0.728-0.848)	-1.859	0.063
GNB	XGBoost	0.734(0.669-0.799)	0.808(0.75-0.866)	-1.986	0.047
KNN	LightGBM	0.703(0.636-0.771)	0.832(0.777-0.887)	-3.184	0.001
KNN	Logit	0.703(0.636-0.771)	0.818(0.761-0.875)	-3.127	0.002
KNN	PLSDA	0.703(0.636-0.771)	0.806(0.748-0.865)	-2.942	0.003
KNN	RF	0.703(0.636-0.771)	0.825(0.769-0.881)	-3.358	0.001
KNN	SVM	0.703(0.636-0.771)	0.788(0.728-0.848)	-2.331	0.020
KNN	XGBoost	0.703(0.636-0.771)	0.808(0.75-0.866)	-2.696	0.007
LightGBM	Logit	0.832(0.777-0.887)	0.818(0.761-0.875)	0.503	0.615
LightGBM	PLSDA	0.832(0.777-0.887)	0.806(0.748-0.865)	0.821	0.412
LightGBM	RF	0.832(0.777-0.887)	0.825(0.769-0.881)	0.352	0.725
LightGBM	SVM	0.832(0.777-0.887)	0.788(0.728-0.848)	1.444	0.149
LightGBM	XGBoost	0.832(0.777-0.887)	0.808(0.75-0.866)	0.885	0.376
Logit	PLSDA	0.818(0.761-0.875)	0.806(0.748-0.865)	0.472	0.637
Logit	**RF**	**0.818(0.761-0.875)**	**0.825(0.769-0.881)**	**-0.272**	**0.786**
Logit	**SVM**	**0.818(0.761-0.875)**	**0.788(0.728-0.848)**	**1.810**	**0.070**
Logit	XGBoost	0.818(0.761-0.875)	0.808(0.75-0.866)	0.313	0.754
PLSDA	RF	0.806(0.748-0.865)	0.825(0.769-0.881)	-0.643	0.520
PLSDA	SVM	0.806(0.748-0.865)	0.788(0.728-0.848)	0.742	0.458
PLSDA	XGBoost	0.806(0.748-0.865)	0.808(0.75-0.866)	-0.049	0.961
RF	**SVM**	**0.825(0.769-0.881)**	**0.788(0.728-0.848)**	**1.305**	**0.192**
RF	XGBoost	0.825(0.769-0.881)	0.808(0.75-0.866)	0.689	0.491
SVM	XGBoost	0.788(0.728-0.848)	0.808(0.75-0.866)	-0.578	0.563

Note: BPNN, Back-Propagation Neural Network; DT, Decision Tree; GBDT, Gradient Boosting Decision Tree; GNB, Gaussian Naïve Bayesian; KNN, k-Nearest Neighbor; LightGBM, Light Gradient Boosting Machine; Logit, Logistic Regression; PLSDA, Partial Least Squares Discriminant Analysis; RF,Random Forest; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 13 Ten-fold cross validation results of different combinations stacking ensemble models

Model	Fold	AUC(95%CI)	Accuracy	F1 score	NPV	PPV	Sensitivity	Specificity
DT+SVM	**0**	**0.915(0.848-0.981)**	**0.930**	**0.902**	**0.915**	**0.958**	**0.852**	**0.977**
DT+SVM	**1**	**0.818(0.728-0.908)**	**0.887**	**0.778**	**0.860**	**1.000**	**0.636**	**1.000**
DT+SVM	**2**	**0.935(0.877-0.994)**	**0.944**	**0.931**	**0.909**	**1.000**	**0.871**	**1.000**
DT+SVM	**3**	**0.881(0.805-0.957)**	**0.930**	**0.865**	**0.909**	**1.000**	**0.762**	**1.000**
DT+SVM	**4**	**0.804(0.711-0.896)**	**0.845**	**0.756**	**0.796**	**1.000**	**0.607**	**1.000**
DT+SVM	**5**	**0.865(0.785-0.945)**	**0.915**	**0.833**	**0.909**	**0.938**	**0.750**	**0.980**
DT+SVM	**6**	**0.891(0.817-0.965)**	**0.929**	**0.878**	**0.904**	**1.000**	**0.783**	**1.000**
DT+SVM	**7**	**0.929(0.867-0.99)**	**0.943**	**0.917**	**0.936**	**0.957**	**0.880**	**0.978**
DT+SVM	**8**	**0.893(0.819-0.966)**	**0.914**	**0.880**	**0.875**	**1.000**	**0.786**	**1.000**
DT+SVM	**9**	**0.82(0.729-0.911)**	**0.871**	**0.780**	**0.833**	**1.000**	**0.640**	**1.000**
DT+SVM+Logit	**0**	**0.866(0.786-0.946)**	**0.887**	**0.840**	**0.875**	**0.913**	**0.778**	**0.955**
DT+SVM+Logit	**1**	**0.841(0.755-0.927)**	**0.901**	**0.811**	**0.875**	**1.000**	**0.682**	**1.000**

DT+SVM+Logit	**2**	**0.907(0.838-0.975)**	**0.915**	**0.897**	**0.886**	**0.963**	**0.839**	**0.975**
DT+SVM+Logit	**3**	**0.857(0.775-0.939)**	**0.915**	**0.833**	**0.893**	**1.000**	**0.714**	**1.000**
DT+SVM+Logit	**4**	**0.762(0.663-0.862)**	**0.803**	**0.696**	**0.774**	**0.889**	**0.571**	**0.953**
DT+SVM+Logit	**5**	**0.83(0.743-0.918)**	**0.887**	**0.778**	**0.891**	**0.875**	**0.700**	**0.961**
DT+SVM+Logit	**6**	**0.849(0.764-0.933)**	**0.871**	**0.800**	**0.896**	**0.818**	**0.783**	**0.915**
DT+SVM+Logit	**7**	**0.876(0.797-0.954)**	**0.886**	**0.840**	**0.911**	**0.840**	**0.840**	**0.911**
DT+SVM+Logit	**8**	**0.863(0.782-0.944)**	**0.886**	**0.840**	**0.854**	**0.955**	**0.750**	**0.976**
DT+SVM+Logit	**9**	**0.829(0.74-0.918)**	**0.871**	**0.791**	**0.846**	**0.944**	**0.680**	**0.978**
DT+SVM+Logit+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
DT+SVM+Logit+RF	1	0.808(0.716-0.9)	0.873	0.757	0.857	0.933	0.636	0.980
DT+SVM+Logit+RF	2	0.891(0.817-0.964)	0.901	0.877	0.867	0.962	0.806	0.975
DT+SVM+Logit+RF	3	0.847(0.763-0.932)	0.901	0.811	0.891	0.938	0.714	0.980
DT+SVM+Logit+RF	4	0.769(0.67-0.867)	0.803	0.708	0.784	0.850	0.607	0.930
DT+SVM+Logit+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
DT+SVM+Logit+RF	6	0.86(0.778-0.942)	0.871	0.809	0.913	0.792	0.826	0.894
DT+SVM+Logit+RF	7	0.864(0.783-0.945)	0.871	0.824	0.909	0.808	0.840	0.889
DT+SVM+Logit+RF	8	0.881(0.804-0.958)	0.900	0.863	0.872	0.957	0.786	0.976
DT+SVM+Logit+RF	9	0.829(0.74-0.918)	0.871	0.791	0.846	0.944	0.680	0.978
DT+SVM+RF	0	0.873(0.795-0.951)	0.887	0.846	0.891	0.880	0.815	0.932
DT+SVM+RF	1	0.818(0.728-0.908)	0.887	0.778	0.860	1.000	0.636	1.000
DT+SVM+RF	2	0.878(0.801-0.955)	0.887	0.862	0.864	0.926	0.806	0.950
DT+SVM+RF	3	0.871(0.792-0.95)	0.915	0.842	0.907	0.941	0.762	0.980
DT+SVM+RF	4	0.756(0.656-0.856)	0.803	0.682	0.764	0.938	0.536	0.977
DT+SVM+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
DT+SVM+RF	6	0.848(0.764-0.933)	0.886	0.810	0.882	0.895	0.739	0.957
DT+SVM+RF	7	0.876(0.797-0.954)	0.886	0.840	0.911	0.840	0.840	0.911
DT+SVM+RF	8	0.893(0.819-0.966)	0.914	0.880	0.875	1.000	0.786	1.000
DT+SVM+RF	9	0.82(0.729-0.911)	0.871	0.780	0.833	1.000	0.640	1.000
DT+Logit	0	0.739(0.637-0.842)	0.775	0.667	0.780	0.762	0.593	0.886
DT+Logit	1	0.777(0.68-0.875)	0.831	0.700	0.849	0.778	0.636	0.918
DT+Logit	2	0.808(0.716-0.901)	0.817	0.780	0.814	0.821	0.742	0.875
DT+Logit	3	0.79(0.694-0.885)	0.859	0.722	0.857	0.867	0.619	0.960
DT+Logit	4	0.686(0.577-0.794)	0.732	0.578	0.722	0.765	0.464	0.907
DT+Logit	5	0.726(0.622-0.83)	0.803	0.611	0.836	0.688	0.550	0.902
DT+Logit	6	0.709(0.602-0.815)	0.743	0.609	0.809	0.609	0.609	0.809
DT+Logit	7	0.773(0.675-0.872)	0.800	0.708	0.830	0.739	0.680	0.867
DT+Logit	8	0.75(0.648-0.852)	0.786	0.681	0.765	0.842	0.571	0.929
DT+Logit	9	0.758(0.657-0.859)	0.814	0.683	0.796	0.875	0.560	0.956
DT+Logit+RF	0	0.843(0.758-0.929)	0.859	0.808	0.870	0.840	0.778	0.909
DT+Logit+RF	1	0.808(0.716-0.9)	0.873	0.757	0.857	0.933	0.636	0.980
DT+Logit+RF	2	0.846(0.761-0.931)	0.859	0.821	0.826	0.920	0.742	0.950
DT+Logit+RF	3	0.847(0.763-0.932)	0.901	0.811	0.891	0.938	0.714	0.980
DT+Logit+RF	4	0.715(0.61-0.82)	0.761	0.622	0.741	0.824	0.500	0.930
DT+Logit+RF	5	0.83(0.743-0.918)	0.887	0.778	0.891	0.875	0.700	0.961
DT+Logit+RF	6	0.795(0.7-0.89)	0.829	0.727	0.857	0.762	0.696	0.894

DT+Logit+RF	7	0.864(0.783-0.945)	0.871	0.824	0.909	0.808	0.840	0.889
DT+Logit+RF	8	0.815(0.724-0.907)	0.843	0.776	0.816	0.905	0.679	0.952
DT+Logit+RF	9	0.789(0.693-0.885)	0.843	0.732	0.815	0.938	0.600	0.978
DT+RF	0	0.832(0.745-0.92)	0.845	0.792	0.867	0.808	0.778	0.886
DT+RF	1	0.798(0.704-0.892)	0.859	0.737	0.855	0.875	0.636	0.959
DT+RF	2	0.862(0.781-0.943)	0.873	0.842	0.844	0.923	0.774	0.950
DT+RF	3	0.861(0.78-0.942)	0.901	0.821	0.906	0.889	0.762	0.960
DT+RF	4	0.756(0.656-0.856)	0.803	0.682	0.764	0.938	0.536	0.977
DT+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
DT+RF	6	0.816(0.725-0.908)	0.843	0.756	0.875	0.773	0.739	0.894
DT+RF	7	0.876(0.797-0.954)	0.886	0.840	0.911	0.840	0.840	0.911
DT+RF	8	0.833(0.745-0.921)	0.857	0.800	0.833	0.909	0.714	0.952
DT+RF	9	0.809(0.716-0.902)	0.857	0.762	0.830	0.941	0.640	0.978
GBDT+DT	0	0.866(0.786-0.946)	0.887	0.840	0.875	0.913	0.778	0.955
GBDT+DT	1	0.831(0.743-0.919)	0.887	0.789	0.873	0.938	0.682	0.980
GBDT+DT	2	0.833(0.746-0.921)	0.845	0.807	0.822	0.885	0.742	0.925
GBDT+DT	3	0.895(0.822-0.967)	0.930	0.872	0.925	0.944	0.810	0.980
GBDT+DT	4	0.733(0.63-0.836)	0.775	0.652	0.755	0.833	0.536	0.930
GBDT+DT	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
GBDT+DT	6	0.87(0.79-0.95)	0.900	0.837	0.900	0.900	0.783	0.957
GBDT+DT	7	0.887(0.812-0.962)	0.900	0.857	0.913	0.875	0.840	0.933
GBDT+DT	8	0.833(0.745-0.921)	0.857	0.800	0.833	0.909	0.714	0.952
GBDT+DT	9	0.809(0.716-0.902)	0.857	0.762	0.830	0.941	0.640	0.978
GBDT+DT+SVM	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+DT+SVM	1	0.841(0.755-0.927)	0.901	0.811	0.875	1.000	0.682	1.000
GBDT+DT+SVM	2	0.862(0.781-0.943)	0.873	0.842	0.844	0.923	0.774	0.950
GBDT+DT+SVM	3	0.905(0.835-0.974)	0.944	0.895	0.926	1.000	0.810	1.000
GBDT+DT+SVM	4	0.756(0.656-0.856)	0.803	0.682	0.764	0.938	0.536	0.977
GBDT+DT+SVM	5	0.865(0.785-0.945)	0.915	0.833	0.909	0.938	0.750	0.980
GBDT+DT+SVM	6	0.892(0.818-0.965)	0.914	0.864	0.918	0.905	0.826	0.957
GBDT+DT+SVM	7	0.887(0.812-0.962)	0.900	0.857	0.913	0.875	0.840	0.933
GBDT+DT+SVM	8	0.863(0.782-0.944)	0.886	0.840	0.854	0.955	0.750	0.976
GBDT+DT+SVM	9	0.809(0.716-0.902)	0.857	0.762	0.830	0.941	0.640	0.978
GBDT+DT+SVM+Logit	**0**	**0.885(0.81-0.96)**	**0.901**	**0.863**	**0.894**	**0.917**	**0.815**	**0.955**
GBDT+DT+SVM+Logit	**1**	**0.831(0.743-0.919)**	**0.887**	**0.789**	**0.873**	**0.938**	**0.682**	**0.980**
GBDT+DT+SVM+Logit	**2**	**0.846(0.761-0.931)**	**0.859**	**0.821**	**0.826**	**0.920**	**0.742**	**0.950**
GBDT+DT+SVM+Logit	**3**	**0.905(0.835-0.974)**	**0.944**	**0.895**	**0.926**	**1.000**	**0.810**	**1.000**
GBDT+DT+SVM+Logit	**4**	**0.757(0.657-0.857)**	**0.789**	**0.694**	**0.780**	**0.810**	**0.607**	**0.907**
GBDT+DT+SVM+Logit	**5**	**0.855(0.773-0.938)**	**0.901**	**0.811**	**0.907**	**0.882**	**0.750**	**0.961**
GBDT+DT+SVM+Logit	**6**	**0.892(0.818-0.965)**	**0.914**	**0.864**	**0.918**	**0.905**	**0.826**	**0.957**
GBDT+DT+SVM+Logit	**7**	**0.864(0.783-0.945)**	**0.871**	**0.824**	**0.909**	**0.808**	**0.840**	**0.889**
GBDT+DT+SVM+Logit	**8**	**0.845(0.76-0.931)**	**0.871**	**0.816**	**0.837**	**0.952**	**0.714**	**0.976**
GBDT+DT+SVM+Logit	**9**	**0.82(0.729-0.911)**	**0.871**	**0.780**	**0.833**	**1.000**	**0.640**	**1.000**
GBDT+DT+SVM+Logit+RF	**0**	**0.873(0.795-0.951)**	**0.887**	**0.846**	**0.891**	**0.880**	**0.815**	**0.932**
GBDT+DT+SVM+Logit+RF	**1**	**0.831(0.743-0.919)**	**0.887**	**0.789**	**0.873**	**0.938**	**0.682**	**0.980**

GBDT+DT+SVM+Logit+RF	**2**	**0.846(0.761-0.931)**	**0.859**	**0.821**	**0.826**	**0.920**	**0.742**	**0.950**
GBDT+DT+SVM+Logit+RF	**3**	**0.881(0.805-0.957)**	**0.930**	**0.865**	**0.909**	**1.000**	**0.762**	**1.000**
GBDT+DT+SVM+Logit+RF	**4**	**0.757(0.657-0.857)**	**0.789**	**0.694**	**0.780**	**0.810**	**0.607**	**0.907**
GBDT+DT+SVM+Logit+RF	**5**	**0.855(0.773-0.938)**	**0.901**	**0.811**	**0.907**	**0.882**	**0.750**	**0.961**
GBDT+DT+SVM+Logit+RF	**6**	**0.859(0.777-0.942)**	**0.886**	**0.818**	**0.898**	**0.857**	**0.783**	**0.936**
GBDT+DT+SVM+Logit+RF	**7**	**0.876(0.797-0.954)**	**0.886**	**0.840**	**0.911**	**0.840**	**0.840**	**0.911**
GBDT+DT+SVM+Logit+RF	**8**	**0.845(0.76-0.931)**	**0.871**	**0.816**	**0.837**	**0.952**	**0.714**	**0.976**
GBDT+DT+SVM+Logit+RF	**9**	**0.82(0.729-0.911)**	**0.871**	**0.780**	**0.833**	**1.000**	**0.640**	**1.000**
GBDT+DT+SVM+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+DT+SVM+RF	1	0.808(0.716-0.9)	0.873	0.757	0.857	0.933	0.636	0.980
GBDT+DT+SVM+RF	2	0.862(0.781-0.943)	0.873	0.842	0.844	0.923	0.774	0.950
GBDT+DT+SVM+RF	3	0.905(0.835-0.974)	0.944	0.895	0.926	1.000	0.810	1.000
GBDT+DT+SVM+RF	4	0.733(0.63-0.836)	0.775	0.652	0.755	0.833	0.536	0.930
GBDT+DT+SVM+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
GBDT+DT+SVM+RF	6	0.892(0.818-0.965)	0.914	0.864	0.918	0.905	0.826	0.957
GBDT+DT+SVM+RF	7	0.887(0.812-0.962)	0.900	0.857	0.913	0.875	0.840	0.933
GBDT+DT+SVM+RF	8	0.851(0.767-0.935)	0.871	0.824	0.851	0.913	0.750	0.952
GBDT+DT+SVM+RF	9	0.809(0.716-0.902)	0.857	0.762	0.830	0.941	0.640	0.978
GBDT+DT+Logit	**0**	**0.855(0.772-0.937)**	**0.873**	**0.824**	**0.872**	**0.875**	**0.778**	**0.932**
GBDT+DT+Logit	**1**	**0.831(0.743-0.919)**	**0.887**	**0.789**	**0.873**	**0.938**	**0.682**	**0.980**
GBDT+DT+Logit	**2**	**0.833(0.746-0.921)**	**0.845**	**0.807**	**0.822**	**0.885**	**0.742**	**0.925**
GBDT+DT+Logit	**3**	**0.871(0.792-0.95)**	**0.915**	**0.842**	**0.907**	**0.941**	**0.762**	**0.980**
GBDT+DT+Logit	**4**	**0.721(0.617-0.826)**	**0.761**	**0.638**	**0.750**	**0.789**	**0.536**	**0.907**
GBDT+DT+Logit	**5**	**0.855(0.773-0.938)**	**0.901**	**0.811**	**0.907**	**0.882**	**0.750**	**0.961**
GBDT+DT+Logit	**6**	**0.838(0.751-0.925)**	**0.871**	**0.791**	**0.880**	**0.850**	**0.739**	**0.936**
GBDT+DT+Logit	**7**	**0.864(0.783-0.945)**	**0.871**	**0.824**	**0.909**	**0.808**	**0.840**	**0.889**
GBDT+DT+Logit	**8**	**0.827(0.738-0.917)**	**0.857**	**0.792**	**0.820**	**0.950**	**0.679**	**0.976**
GBDT+DT+Logit	**9**	**0.798(0.703-0.892)**	**0.843**	**0.744**	**0.827**	**0.889**	**0.640**	**0.956**
GBDT+DT+Logit+RF	**0**	**0.873(0.795-0.951)**	**0.887**	**0.846**	**0.891**	**0.880**	**0.815**	**0.932**
GBDT+DT+Logit+RF	**1**	**0.831(0.743-0.919)**	**0.887**	**0.789**	**0.873**	**0.938**	**0.682**	**0.980**
GBDT+DT+Logit+RF	**2**	**0.846(0.761-0.931)**	**0.859**	**0.821**	**0.826**	**0.920**	**0.742**	**0.950**
GBDT+DT+Logit+RF	**3**	**0.881(0.805-0.957)**	**0.930**	**0.865**	**0.909**	**1.000**	**0.762**	**1.000**
GBDT+DT+Logit+RF	**4**	**0.757(0.657-0.857)**	**0.789**	**0.694**	**0.780**	**0.810**	**0.607**	**0.907**
GBDT+DT+Logit+RF	**5**	**0.855(0.773-0.938)**	**0.901**	**0.811**	**0.907**	**0.882**	**0.750**	**0.961**
GBDT+DT+Logit+RF	**6**	**0.859(0.777-0.942)**	**0.886**	**0.818**	**0.898**	**0.857**	**0.783**	**0.936**
GBDT+DT+Logit+RF	**7**	**0.876(0.797-0.954)**	**0.886**	**0.840**	**0.911**	**0.840**	**0.840**	**0.911**
GBDT+DT+Logit+RF	**8**	**0.845(0.76-0.931)**	**0.871**	**0.816**	**0.837**	**0.952**	**0.714**	**0.976**
GBDT+DT+Logit+RF	**9**	**0.82(0.729-0.911)**	**0.871**	**0.780**	**0.833**	**1.000**	**0.640**	**1.000**
GBDT+DT+RF	0	0.873(0.795-0.951)	0.887	0.846	0.891	0.880	0.815	0.932
GBDT+DT+RF	1	0.808(0.716-0.9)	0.873	0.757	0.857	0.933	0.636	0.980
GBDT+DT+RF	2	0.833(0.746-0.921)	0.845	0.807	0.822	0.885	0.742	0.925
GBDT+DT+RF	3	0.895(0.822-0.967)	0.930	0.872	0.925	0.944	0.810	0.980
GBDT+DT+RF	4	0.733(0.63-0.836)	0.775	0.652	0.755	0.833	0.536	0.930
GBDT+DT+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
GBDT+DT+RF	6	0.892(0.818-0.965)	0.914	0.864	0.918	0.905	0.826	0.957

GBDT+DT+RF	7	0.887(0.812-0.962)	0.900	0.857	0.913	0.875	0.840	0.933
GBDT+DT+RF	8	0.851(0.767-0.935)	0.871	0.824	0.851	0.913	0.750	0.952
GBDT+DT+RF	9	0.809(0.716-0.902)	0.857	0.762	0.830	0.941	0.640	0.978
GBDT+SVM	0	0.907(0.839-0.976)	0.930	0.898	0.898	1.000	0.815	1.000
GBDT+SVM	1	0.909(0.841-0.977)	0.944	0.900	0.925	1.000	0.818	1.000
GBDT+SVM	2	0.919(0.855-0.984)	0.930	0.912	0.889	1.000	0.839	1.000
GBDT+SVM	3	0.952(0.901-1.004)	0.972	0.950	0.962	1.000	0.905	1.000
GBDT+SVM	4	0.881(0.805-0.957)	0.901	0.863	0.875	0.957	0.786	0.977
GBDT+SVM	5	0.9(0.829-0.971)	0.944	0.889	0.927	1.000	0.800	1.000
GBDT+SVM	6	0.946(0.891-1)	0.957	0.933	0.958	0.955	0.913	0.979
GBDT+SVM	7	0.96(0.912-1.008)	0.971	0.958	0.957	1.000	0.920	1.000
GBDT+SVM	8	0.899(0.827-0.97)	0.914	0.885	0.891	0.958	0.821	0.976
GBDT+SVM	9	0.88(0.803-0.957)	0.914	0.864	0.882	1.000	0.760	1.000
GBDT+SVM+Logit	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+SVM+Logit	1	0.841(0.755-0.927)	0.901	0.811	0.875	1.000	0.682	1.000
GBDT+SVM+Logit	2	0.875(0.797-0.952)	0.887	0.857	0.848	0.960	0.774	0.975
GBDT+SVM+Logit	3	0.929(0.867-0.99)	0.958	0.923	0.943	1.000	0.857	1.000
GBDT+SVM+Logit	4	0.804(0.712-0.897)	0.831	0.760	0.816	0.864	0.679	0.930
GBDT+SVM+Logit	5	0.88(0.804-0.957)	0.915	0.842	0.925	0.889	0.800	0.961
GBDT+SVM+Logit	6	0.892(0.818-0.965)	0.914	0.864	0.918	0.905	0.826	0.957
GBDT+SVM+Logit	7	0.898(0.826-0.97)	0.914	0.875	0.915	0.913	0.840	0.956
GBDT+SVM+Logit	8	0.845(0.76-0.931)	0.871	0.816	0.837	0.952	0.714	0.976
GBDT+SVM+Logit	9	0.84(0.753-0.927)	0.886	0.810	0.849	1.000	0.680	1.000
GBDT+SVM+Logit+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+SVM+Logit+RF	1	0.831(0.743-0.919)	0.887	0.789	0.873	0.938	0.682	0.980
GBDT+SVM+Logit+RF	2	0.846(0.761-0.931)	0.859	0.821	0.826	0.920	0.742	0.950
GBDT+SVM+Logit+RF	3	0.905(0.835-0.974)	0.944	0.895	0.926	1.000	0.810	1.000
GBDT+SVM+Logit+RF	4	0.757(0.657-0.857)	0.789	0.694	0.780	0.810	0.607	0.907
GBDT+SVM+Logit+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
GBDT+SVM+Logit+RF	6	0.87(0.79-0.95)	0.900	0.837	0.900	0.900	0.783	0.957
GBDT+SVM+Logit+RF	7	0.876(0.797-0.954)	0.886	0.840	0.911	0.840	0.840	0.911
GBDT+SVM+Logit+RF	8	0.845(0.76-0.931)	0.871	0.816	0.837	0.952	0.714	0.976
GBDT+SVM+Logit+RF	9	0.82(0.729-0.911)	0.871	0.780	0.833	1.000	0.640	1.000
GBDT+SVM+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+SVM+RF	1	0.841(0.755-0.927)	0.901	0.811	0.875	1.000	0.682	1.000
GBDT+SVM+RF	2	0.862(0.781-0.943)	0.873	0.842	0.844	0.923	0.774	0.950
GBDT+SVM+RF	3	0.929(0.867-0.99)	0.958	0.923	0.943	1.000	0.857	1.000
GBDT+SVM+RF	4	0.852(0.768-0.935)	0.873	0.824	0.854	0.913	0.750	0.953
GBDT+SVM+RF	5	0.9(0.829-0.971)	0.944	0.889	0.927	1.000	0.800	1.000
GBDT+SVM+RF	6	0.914(0.847-0.98)	0.929	0.889	0.938	0.909	0.870	0.957
GBDT+SVM+RF	7	0.907(0.837-0.976)	0.914	0.880	0.933	0.880	0.880	0.933
GBDT+SVM+RF	8	0.863(0.782-0.944)	0.886	0.840	0.854	0.955	0.750	0.976
GBDT+SVM+RF	9	0.84(0.753-0.927)	0.886	0.810	0.849	1.000	0.680	1.000
GBDT+Logit	0	0.866(0.786-0.946)	0.887	0.840	0.875	0.913	0.778	0.955
GBDT+Logit	1	0.831(0.743-0.919)	0.887	0.789	0.873	0.938	0.682	0.980

GBDT+Logit	2	0.846(0.761-0.931)	0.859	0.821	0.826	0.920	0.742	0.950
GBDT+Logit	3	0.857(0.775-0.939)	0.915	0.833	0.893	1.000	0.714	1.000
GBDT+Logit	4	0.775(0.677-0.872)	0.803	0.720	0.796	0.818	0.643	0.907
GBDT+Logit	5	0.821(0.731-0.91)	0.873	0.757	0.889	0.824	0.700	0.941
GBDT+Logit	6	0.848(0.764-0.933)	0.886	0.810	0.882	0.895	0.739	0.957
GBDT+Logit	7	0.887(0.812-0.962)	0.900	0.857	0.913	0.875	0.840	0.933
GBDT+Logit	8	0.827(0.738-0.917)	0.857	0.792	0.820	0.950	0.679	0.976
GBDT+Logit	9	0.82(0.729-0.911)	0.871	0.780	0.833	1.000	0.640	1.000
GBDT+Logit+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+Logit+RF	1	0.831(0.743-0.919)	0.887	0.789	0.873	0.938	0.682	0.980
GBDT+Logit+RF	2	0.846(0.761-0.931)	0.859	0.821	0.826	0.920	0.742	0.950
GBDT+Logit+RF	3	0.881(0.805-0.957)	0.930	0.865	0.909	1.000	0.762	1.000
GBDT+Logit+RF	4	0.775(0.677-0.872)	0.803	0.720	0.796	0.818	0.643	0.907
GBDT+Logit+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
GBDT+Logit+RF	6	0.87(0.79-0.95)	0.900	0.837	0.900	0.900	0.783	0.957
GBDT+Logit+RF	7	0.876(0.797-0.954)	0.886	0.840	0.911	0.840	0.840	0.911
GBDT+Logit+RF	8	0.845(0.76-0.931)	0.871	0.816	0.837	0.952	0.714	0.976
GBDT+Logit+RF	9	0.82(0.729-0.911)	0.871	0.780	0.833	1.000	0.640	1.000
GBDT+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
GBDT+RF	1	0.841(0.755-0.927)	0.901	0.811	0.875	1.000	0.682	1.000
GBDT+RF	2	0.862(0.781-0.943)	0.873	0.842	0.844	0.923	0.774	0.950
GBDT+RF	3	0.929(0.867-0.99)	0.958	0.923	0.943	1.000	0.857	1.000
GBDT+RF	4	0.834(0.747-0.921)	0.859	0.800	0.837	0.909	0.714	0.953
GBDT+RF	5	0.9(0.829-0.971)	0.944	0.889	0.927	1.000	0.800	1.000
GBDT+RF	6	0.914(0.847-0.98)	0.929	0.889	0.938	0.909	0.870	0.957
GBDT+RF	7	0.907(0.837-0.976)	0.914	0.880	0.933	0.880	0.880	0.933
GBDT+RF	8	0.857(0.774-0.94)	0.886	0.833	0.840	1.000	0.714	1.000
GBDT+RF	9	0.84(0.753-0.927)	0.886	0.810	0.849	1.000	0.680	1.000
SVM+Logit	0	0.97(0.928-1.012)	0.972	0.963	0.977	0.963	0.963	0.977
SVM+Logit	1	0.864(0.783-0.944)	0.915	0.842	0.891	1.000	0.727	1.000
SVM+Logit	2	0.939(0.882-0.996)	0.944	0.933	0.929	0.966	0.903	0.975
SVM+Logit	3	0.929(0.867-0.99)	0.958	0.923	0.943	1.000	0.857	1.000
SVM+Logit	4	0.863(0.783-0.944)	0.887	0.840	0.857	0.955	0.750	0.977
SVM+Logit	5	0.94(0.884-0.997)	0.958	0.923	0.962	0.947	0.900	0.980
SVM+Logit	6	0.924(0.861-0.987)	0.943	0.909	0.939	0.952	0.870	0.979
SVM+Logit	7	0.927(0.864-0.989)	0.929	0.902	0.955	0.885	0.920	0.933
SVM+Logit	8	0.917(0.851-0.983)	0.929	0.906	0.911	0.960	0.857	0.976
SVM+Logit	9	0.86(0.778-0.942)	0.900	0.837	0.865	1.000	0.720	1.000
SVM+Logit+RF	0	0.903(0.833-0.973)	0.915	0.885	0.913	0.920	0.852	0.955
SVM+Logit+RF	1	0.841(0.755-0.927)	0.901	0.811	0.875	1.000	0.682	1.000
SVM+Logit+RF	2	0.907(0.838-0.975)	0.915	0.897	0.886	0.963	0.839	0.975
SVM+Logit+RF	3	0.881(0.805-0.957)	0.930	0.865	0.909	1.000	0.762	1.000
SVM+Logit+RF	4	0.78(0.684-0.877)	0.817	0.723	0.788	0.895	0.607	0.953
SVM+Logit+RF	5	0.855(0.773-0.938)	0.901	0.811	0.907	0.882	0.750	0.961
SVM+Logit+RF	6	0.859(0.777-0.942)	0.886	0.818	0.898	0.857	0.783	0.936

SVM+Logit+RF	7	0.884(0.809-0.96)	0.886	0.846	0.930	0.815	0.880	0.889
SVM+Logit+RF	8	0.863(0.782-0.944)	0.886	0.840	0.854	0.955	0.750	0.976
SVM+Logit+RF	9	0.84(0.753-0.927)	0.886	0.810	0.849	1.000	0.680	1.000
SVM+RF	0	0.926(0.864-0.988)	0.944	0.920	0.917	1.000	0.852	1.000
SVM+RF	1	0.841(0.755-0.927)	0.901	0.811	0.875	1.000	0.682	1.000
SVM+RF	2	0.919(0.855-0.984)	0.930	0.912	0.889	1.000	0.839	1.000
SVM+RF	3	0.952(0.901-1.004)	0.972	0.950	0.962	1.000	0.905	1.000
SVM+RF	4	0.881(0.805-0.957)	0.901	0.863	0.875	0.957	0.786	0.977
SVM+RF	5	0.89(0.817-0.964)	0.930	0.865	0.926	0.941	0.800	0.980
SVM+RF	6	0.924(0.861-0.987)	0.943	0.909	0.939	0.952	0.870	0.979
SVM+RF	7	0.918(0.852-0.983)	0.929	0.898	0.935	0.917	0.880	0.956
SVM+RF	8	0.929(0.867-0.99)	0.943	0.923	0.913	1.000	0.857	1.000
SVM+RF	9	0.86(0.778-0.942)	0.900	0.837	0.865	1.000	0.720	1.000
Logit+RF	0	0.885(0.81-0.96)	0.901	0.863	0.894	0.917	0.815	0.955
Logit+RF	1	0.831(0.743-0.919)	0.887	0.789	0.873	0.938	0.682	0.980
Logit+RF	2	0.846(0.761-0.931)	0.859	0.821	0.826	0.920	0.742	0.950
Logit+RF	3	0.847(0.763-0.932)	0.901	0.811	0.891	0.938	0.714	0.980
Logit+RF	4	0.769(0.67-0.867)	0.803	0.708	0.784	0.850	0.607	0.930
Logit+RF	5	0.83(0.743-0.918)	0.887	0.778	0.891	0.875	0.700	0.961
Logit+RF	6	0.849(0.764-0.933)	0.871	0.800	0.896	0.818	0.783	0.915
Logit+RF	7	0.864(0.783-0.945)	0.871	0.824	0.909	0.808	0.840	0.889
Logit+RF	8	0.845(0.76-0.931)	0.871	0.816	0.837	0.952	0.714	0.976
Logit+RF	9	0.84(0.753-0.927)	0.886	0.810	0.849	1.000	0.680	1.000

Note: Logit: Logistic Regression; RF: Random Forest; DT: Decision Tree; SVM: Support Vector Machine; GBDT, Gradient Boosting Decision Tree; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 14 The comprehensive rank of top 5 independent ML models and various stacking ensemble combinations

Model	AUC ranks	Accuracy ranks	F1 score ranks	G mean ranks	McNemar ranks	NPV ranks	PPV ranks	Sensitivity ranks	Specificity ranks	Comprehensive Rank
DT+SVM	2	2	2	2	23	4	6	9	6	6.222
GBDT+DT+Logit+RF	1	1	1	1	34	3	1	15	5	6.889
DT+SVM+Logit	3	3	3	3	24	5	7	10	7	7.222
GBDT+DT+SVM+Logit	4	7	4	4	25	6	8	12	10	8.889
GBDT+DT+Logit	5	9	5	5	26	7	9	14	11	10.111
DT+SVM+Logit+RF	6	4	6	10	35	13	2	18	1	10.556
GBDT+DT+SVM	10	11	10	6	14	8	14	11	16	11.111
DT+SVM+RF	7	5	7	11	36	14	3	19	2	11.556
GBDT+DT+SVM+RF	11	12	11	7	15	9	15	13	17	12.222
DT+RF	8	6	8	12	37	15	4	20	3	12.556
GBDT+DT+RF	12	13	12	8	16	10	16	16	18	13.444
GBDT+DT+SVM+Logit+RF	9	8	9	13	38	16	5	22	4	13.778
SVM+RF	13	15	13	9	17	11	17	17	24	15.111
GBDT+DT	14	10	14	14	28	17	10	21	9	15.222
SVM+Logit+RF	15	14	15	15	29	18	11	28	12	17.444
GBDT+SVM+Logit	16	17	16	16	18	21	18	23	19	18.222
GBDT+SVM+Logit+RF	17	18	17	17	19	22	19	24	20	19.222
GBDT+SVM+RF	18	19	18	18	20	23	20	25	21	20.222
DT+Logit+RF	21	16	21	21	32	28	12	29	8	20.889
GBDT	24	25	24	24	7	19	29	7	29	20.889
GBDT+Logit+RF	19	20	19	19	21	24	21	26	22	21.222
Logit	28	34	30	26	5	1	35	2	37	22.000
GBDT+RF	20	21	20	20	22	25	22	27	23	22.222
Random Forest	25	29	26	25	6	26	30	8	30	22.778
Logit+RF	22	22	22	22	33	29	13	30	13	22.889
DT	29	32	28	28	3	27	32	6	31	24.000
DT+Logit	26	23	25	29	31	31	23	32	15	26.111
SVM	35	36	35	35	9	20	37	4	39	27.778
GBDT+Logit	27	24	27	30	27	32	25	33	26	27.889
GBDT+SVM	31	27	29	31	30	35	26	35	25	29.889
SVM+Logit	34	31	34	33	13	34	28	34	28	29.889

Note: Logit: Logistic Regression; RF: Random Forest; DT: Decision Tree; SVM: Support Vector Machine; GBDT, Gradient Boosting Decision Tree; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 15 Results of Delong test of the top 10 stacking ensemble ML models

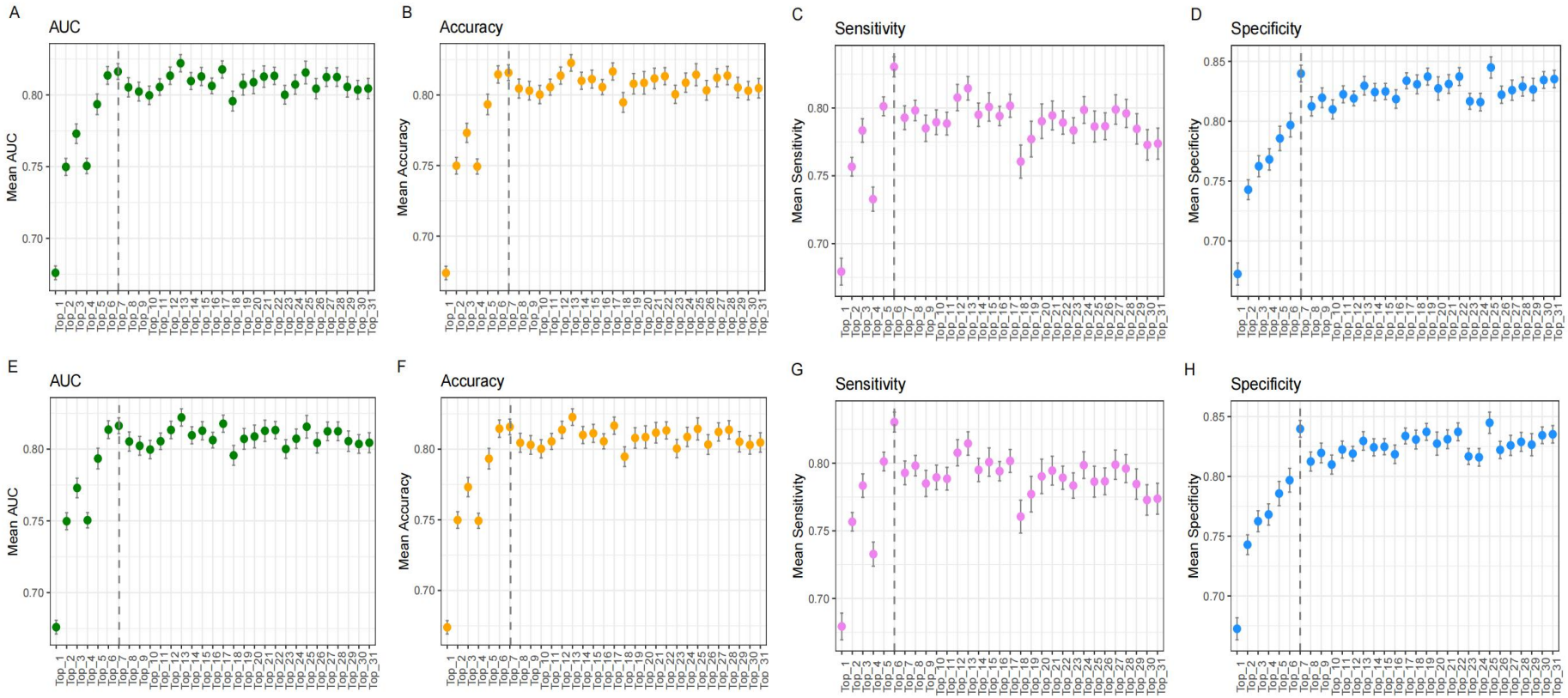
Method 1	Method 2	AUC(95%CI) of Method 1	AUC(95%CI) of Method 2	z	P
DT+RF	DT+SVM	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+RF	DT+SVM+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+RF	DT+SVM+Logit+RF	0.856(0.804-0.908)	0.856(0.804-0.908)	0.000	1.000
DT+RF	DT+SVM+RF	0.856(0.804-0.908)	0.856(0.804-0.908)	0.000	1.000
DT+RF	GBDT+DT+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+RF	GBDT+DT+Logit+RF	0.856(0.804-0.908)	0.864(0.814-0.915)	-1.000	0.317
DT+RF	GBDT+DT+SVM	0.856(0.804-0.908)	0.855(0.803-0.907)	0.083	0.934
DT+RF	GBDT+DT+SVM+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+RF	GBDT+DT+SVM+RF	0.856(0.804-0.908)	0.855(0.803-0.907)	0.083	0.934
DT+SVM	**DT+SVM+Logit**	**0.860(0.809-0.911)**	**0.86(0.809-0.911)**	**0.000**	**1.000**
DT+SVM	DT+SVM+Logit+RF	0.86(0.809-0.911)	0.856(0.804-0.908)	0.392	0.695
DT+SVM	DT+SVM+RF	0.86(0.809-0.911)	0.856(0.804-0.908)	0.392	0.695
DT+SVM	**GBDT+DT+Logit**	**0.86(0.809-0.911)**	**0.86(0.809-0.911)**	**0.000**	**1.000**
DT+SVM	**GBDT+DT+Logit+RF**	**0.86(0.809-0.911)**	**0.864(0.814-0.915)**	**-1.000**	**0.317**
DT+SVM	GBDT+DT+SVM	0.86(0.809-0.911)	0.855(0.803-0.907)	0.576	0.565
DT+SVM	**GBDT+DT+SVM+Logit**	**0.86(0.809-0.911)**	**0.86(0.809-0.911)**	**0.000**	**1.000**
DT+SVM	GBDT+DT+SVM+RF	0.86(0.809-0.911)	0.855(0.803-0.907)	0.576	0.565
DT+SVM+Logit	DT+SVM+Logit+RF	0.86(0.809-0.911)	0.856(0.804-0.908)	0.392	0.695
DT+SVM+Logit	DT+SVM+RF	0.86(0.809-0.911)	0.856(0.804-0.908)	0.392	0.695
DT+SVM+Logit	**GBDT+DT+Logit**	**0.86(0.809-0.911)**	**0.86(0.809-0.911)**	**0.000**	**1.000**
DT+SVM+Logit	GBDT+DT+Logit+RF	0.86(0.809-0.911)	0.864(0.814-0.915)	-1.000	0.317
DT+SVM+Logit	GBDT+DT+SVM	0.86(0.809-0.911)	0.855(0.803-0.907)	0.576	0.565
DT+SVM+Logit	**GBDT+DT+SVM+Logit**	**0.86(0.809-0.911)**	**0.86(0.809-0.911)**	**0.000**	**1.000**
DT+SVM+Logit	**GBDT+DT+SVM+RF**	**0.86(0.809-0.911)**	**0.855(0.803-0.907)**	**0.576**	**0.565**
DT+SVM+Logit+RF	DT+SVM+RF	0.856(0.804-0.908)	0.856(0.804-0.908)	0.000	1.000
DT+SVM+Logit+RF	GBDT+DT+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+SVM+Logit+RF	GBDT+DT+Logit+RF	0.856(0.804-0.908)	0.864(0.814-0.915)	-1.000	0.317
DT+SVM+Logit+RF	GBDT+DT+SVM	0.856(0.804-0.908)	0.855(0.803-0.907)	0.083	0.934
DT+SVM+Logit+RF	GBDT+DT+SVM+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+SVM+Logit+RF	GBDT+DT+SVM+RF	0.856(0.804-0.908)	0.855(0.803-0.907)	0.083	0.934
DT+SVM+RF	GBDT+DT+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+SVM+RF	GBDT+DT+Logit+RF	0.856(0.804-0.908)	0.864(0.814-0.915)	-1.000	0.317
DT+SVM+RF	GBDT+DT+SVM	0.856(0.804-0.908)	0.855(0.803-0.907)	0.083	0.934
DT+SVM+RF	GBDT+DT+SVM+Logit	0.856(0.804-0.908)	0.86(0.809-0.911)	-0.392	0.695
DT+SVM+RF	GBDT+DT+SVM+RF	0.856(0.804-0.908)	0.855(0.803-0.907)	0.083	0.934
GBDT+DT+Logit	**GBDT+DT+Logit+RF**	**0.86(0.809-0.911)**	**0.864(0.814-0.915)**	**-1.000**	**0.317**
GBDT+DT+Logit	GBDT+DT+SVM	0.86(0.809-0.911)	0.855(0.803-0.907)	1.000	0.317
GBDT+DT+Logit	**GBDT+DT+SVM+Logit**	**0.86(0.809-0.911)**	**0.86(0.809-0.911)**	**0.000**	**1.000**
GBDT+DT+Logit	GBDT+DT+SVM+RF	0.86(0.809-0.911)	0.855(0.803-0.907)	1.000	0.317
GBDT+DT+Logit+RF	GBDT+DT+SVM	0.864(0.814-0.915)	0.855(0.803-0.907)	1.421	0.155

GBDT+DT+Logit+RF	**GBDT+DT+SVM+Logit**	**0.864(0.814-0.915)**	**0.86(0.809-0.911)**	**1.000**	**0.317**
GBDT+DT+Logit+RF	GBDT+DT+SVM+RF	0.864(0.814-0.915)	0.855(0.803-0.907)	1.421	0.155
GBDT+DT+SVM	GBDT+DT+SVM+Logit	0.855(0.803-0.907)	0.86(0.809-0.911)	-1.000	0.317
GBDT+DT+SVM	GBDT+DT+SVM+RF	0.855(0.803-0.907)	0.855(0.803-0.907)	0.000	1.000
GBDT+DT+SVM+Logit	GBDT+DT+SVM+RF	0.86(0.809-0.911)	0.855(0.803-0.907)	1.000	0.317

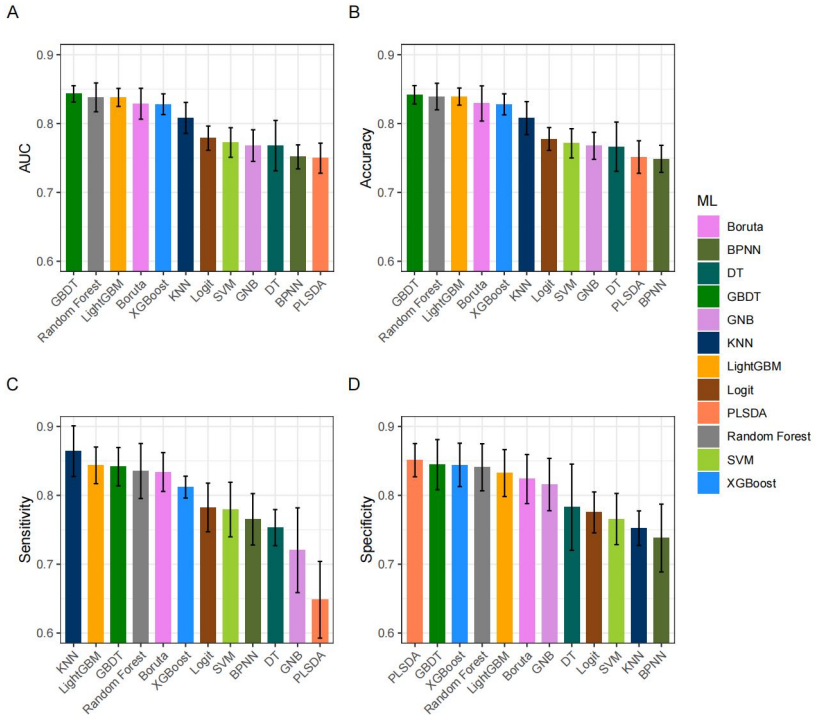
Note: Logit: Logistic Regression; RF: Random Forest; DT: Decision Tree; SVM: Support Vector Machine; GBDT, Gradient Boosting Decision Tree; PPV: positive predictive value; NPV: negative predictive value; AUC: the area under the receiver operating characteristic (ROC) curve.

Supplementary Table 16 Results of the DeLong test comparing the top five independent ML models with the top-performing stacking ensemble ML model

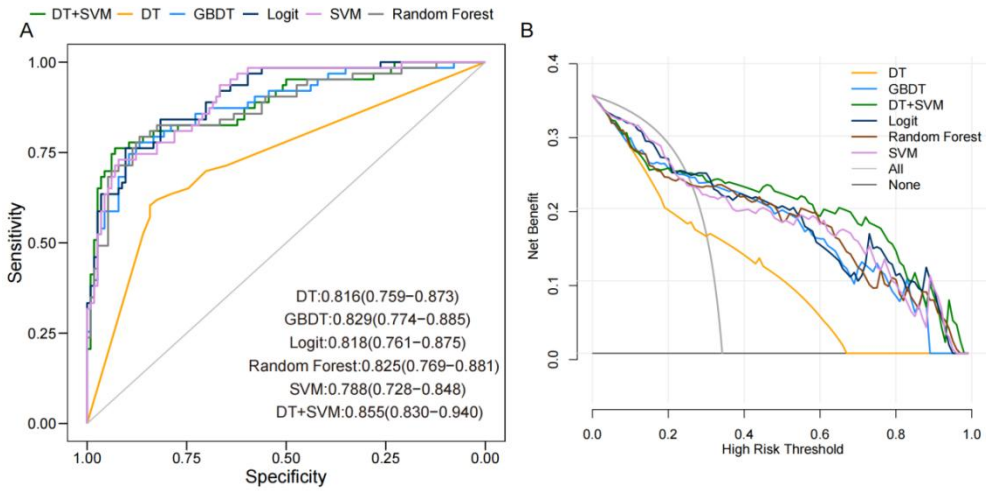
Method_1	Method_2	Method_1	Method_2	z	P
DT+SVM	DT	0.86(0.809-0.911)	0.816(0.759-0.873)	1.626	0.104
DT+SVM	GBDT	0.86(0.809-0.911)	0.829(0.774-0.885)	1.568	0.117
DT+SVM	Logit	0.86(0.809-0.911)	0.818(0.761-0.875)	1.768	0.077
DT+SVM	Random Forest	0.86(0.809-0.911)	0.825(0.769-0.881)	2.126	0.033
DT+SVM	SVM	0.86(0.809-0.911)	0.788(0.728-0.848)	2.716	0.007



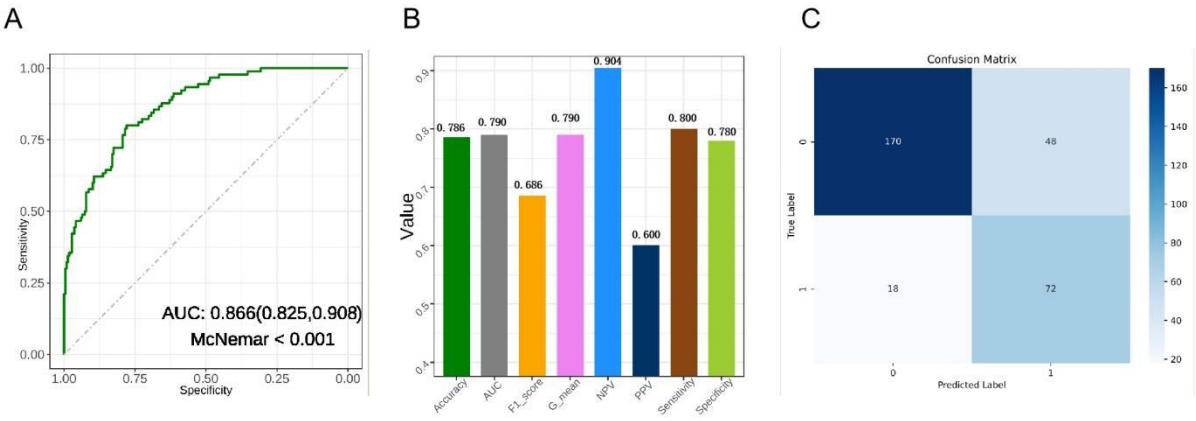
Supplementary Figure 1 The Impact of Different Feature Sets on Model Performance Under Recursive Feature Elimination (RFE). (A) the mean AUC and its 95%CI in training set; (B) the mean accuracy and its 95% CI in training set; (C) the mean sensitivity and its 95% CI in training set; (D) the mean specificity and its 95% CI in training set; (E) the mean AUC and its 95%CI in internal validation set; (F) the mean accuracy and its 95% CI in internal validation set; (G) the mean sensitivity and its 95% CI in internal validation set; (H) the mean specificity and its 95% CI in internal validation set.



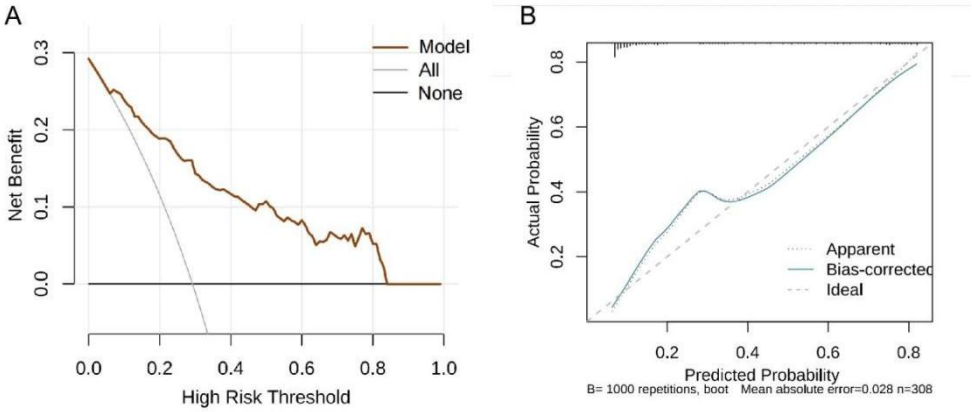
Supplementary Figure 2 The performance of 10-Fold Cross-Validation for 12 Machine Learning Models (A) AUC, (B) Accuracy, (C) Sensitivity, (D) Specificity. Logit: Logistic Regression; PBNN: Pruning Bayesian Neural Network; LightGBM: Light Gradient Boosting Machine; BPNN: Back-Propagation Neural Network; RF: Random Forest; GNB: Gaussian Naive Bayesian; KNN: k-Nearest Neighbor; DT: Decision Tree; SVM: Support Vector Machine; XGBoost: Extreme Gradient Boosting.



Supplementary Figure 3 The ROC and DCA curve performance of top 1 stacking ensemble model and top 5 independent ML models. (A) The ROC curve performance of top 1 stacking ensemble model and top 5 independent ML models; (B) The ROC and DCA curve performance of top 1 stacking ensemble model and top 5 independent ML models. Logit: Logistic Regression; RF: Random Forest; DT: Decision Tree; GBDT, Gradient Boosting Decision Tree; SVM: Support Vector Machine



Supplementary Figure 4 The performance of DT+SVM stacking ML model for prospective validation. (A)Receiver Operating Characteristic Curves of Score for DT+SVM stacking ML model in prospective validation; (B) the performance metric of prospective validation; (C) confusion matrix of prospective validation



Supplementary Figure 5 The DCA and calibration curve performance of DT+SVM stacking ML model for prospective validation. (A) DCA curve; (B) calibration curve; DCA, decision curve analysis.